

Learn Onshape for FTC

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Update Title page with links and ISBN

This book is dedicated to:

My wife who after suffering through my first two books
encouraged me to write this one.

My FTC team that excites me about teaching

My father who spent many hours with me on the Vic 20,
Commodore 64, and the robotic arm science project.
Without his investment, I wouldn't be the engineer I am
today.

FTC coaches everywhere that teach their students to think
well and work hard

*Whatever you do, work at it with all your heart, as working
for the Lord, not for men.
Colossians 3:23 (NIV 1984)*

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Part I.

Getting Started

1. Introduction

This is a little dangerous, because unlike my other books on software where I worked on software professionally for many years, I have never been a professional mechanical engineer.

However, I still feel like many rookie teams could be helped with an introduction to Onshape. (an online CAD system) and specifically with how to use it well for FTC.

There are many excellent resources for Onshape, but a lot of them focus on many details instead of getting you to make useful things quickly. After you have gone through this book, I suspect many students will then go on to take several of the excellent courses at <http://learn.onshape.com>. But many students need quick “wins” and ways they see this is useful before they are ready for all of the detail there.

1.1. Why use CAD (and what is it)?

CAD stands for computer aided design. It allows us to put together designs and show them to our team mates. It falls in between making a drawing on a whiteboard or a piece of paper and actually building the robot.

It provides several significant benefits over a simple drawing.

1. It forces you to make sure that things can actually fit together. (One of the most important principles of robotics is that no two objects can occupy the same space at the same time. - The fancy name for this is impenetrability.¹)
2. You can use part of your design to create 3d printed parts or to send places for fabrication (or use yourself with a laser or a CNC machine) or to even print parts to bend sheet metal for.
3. It is faster than building for figuring out if an idea will work

¹yes, you can argue why this is not “technically” true. But for our purposes it is true enough. As my best friend says, “All education is a series of slightly less egregious lies.”

1. Introduction

4. You can add motion in to make sure things are possible to move. (If it works in CAD, it doesn't mean it will work in the real world. But if it doesn't work in CAD, it is highly unlikely to work in the real world.)
5. You can save versions of the design and compare them.

1.2. Why Onshape?

There are many CAD systems available and many of them are quite good. Why should you choose Onshape over Creo or Solidworks or Fusion 360 or many many others?

1. It runs completely in your browser which means that all installation headaches are gone. I can't overemphasize how important this is.
2. It is like Google Docs for CAD where it was designed from the ground up for collaboration and working on designs with other people. (which makes it great for FTC teams)
3. Many (most?) of the parts commonly used for FTC are already available to find easily thanks to the great work by Purple Gears (FTC #2901)
4. It is a solid choice that can accomplish most of what you need in a CAD system
5. They have excellent training materials and documentation available. (With this point, you might be asking why I am writing this book. I am writing it to help people with the small subset that is necessary and some good patterns. I have also found that many students give up on the training materials because of the amount of depth they go into.)
6. PTC generously offers a free educational version to all students on FTC teams. (Please do not take advantage of this to design commercial products.) . I have a note from PTC's legal team confirming that this covers students that are homeschooled as well as coaches and mentors working with the team.

1.3. Roadmap for our time together

This book is split into 3 main parts.

[Part I: Getting Started](#)

This is the part you are in now, it explains briefly what we are doing and why as well as how to set things up for success. If you are a coach make sure you read [chapter 2](#).

[Part II: Modeling a simple Part](#)

This part focuses on modeling individual parts that you might 3d print.

[Part III: Modeling your robot](#)

This part focuses on assembling parts together (both your own and standard ones)

I recommend you start at the beginning and follow along. I think that doing the exercises will be the best way to learn.

1.4. Get your educational account

You should get an educational account. Some people will use the hobbyist version, and that can be fine but it doesn't have all the power of the educational account. To get an educational account go to <https://onshape.com/edu>

1.5. Setup your account

Go to the upper right where it will have your name. Press the down arrow and select "My account". Under Profile you can change the image that represents you (helpful to your team mates), select a Nickname, etc.

On the left hand side, select "Preferences", scroll down to Units and change "Length Default unit" to millimeter. (Both REV and GoBilda are based on metric so this will make your life easier.) . Be aware that this only changes new documents. (I describe how to change existing documents in [Appendix B](#))

2. For coaches...

Get an educator account, and create a classroom for your team. That way there is a common place that all of your students on the team can put the design documents. I recommend making a folder for each year (for example BioBuzz 2026-2027).

Personally while I use the classroom, I don't setup different classes per year.

Put here directions on how to make a classroom

2.1. “Hudson Bay” Start

Many years ago there was a fur trading company in Canada called the Hudson Bay Company. When they were going on an expedition, they would start by going 1 mile the first day and setting up camp. Why do this when there was so much they needed to do??

Simply, it let them find all of the things they had forgotten when it wasn't too far to go back. And since it was a real start and a real camp setup, it found things that would not have been found with any number of checklists.

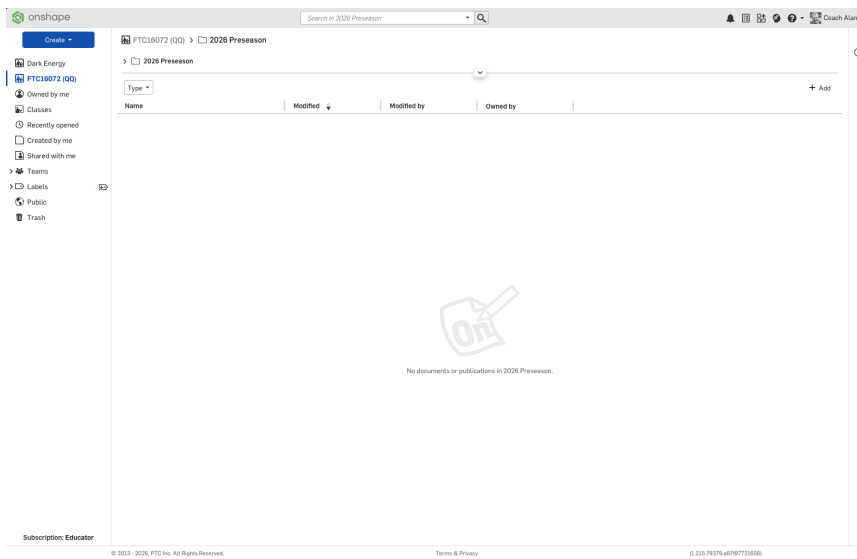
For my teams, our “Hudson Bay” start is that we have every student make a tile with their name on it in a document and then attach it to all of the other tiles. This doesn't show that students have all of the CAD skills necessary, but it shows that their login works and they are able to do a few VERY common things.

So let's begin with how to do this.

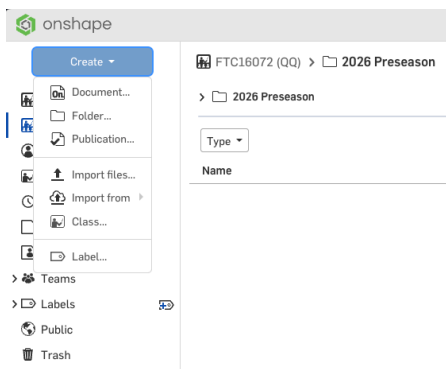
2.2. Create a “team tiles” document

I start each year by creating a new folder that is the year and preseason. (For example, you can see here I did 2026 Preseason)

2. For coaches...



In order to do that, you click on the blue “Create” button in the upper left and then select folder and type in “2026 Preseason”. When you click on the “Create” button, you get a menu that looks like this:

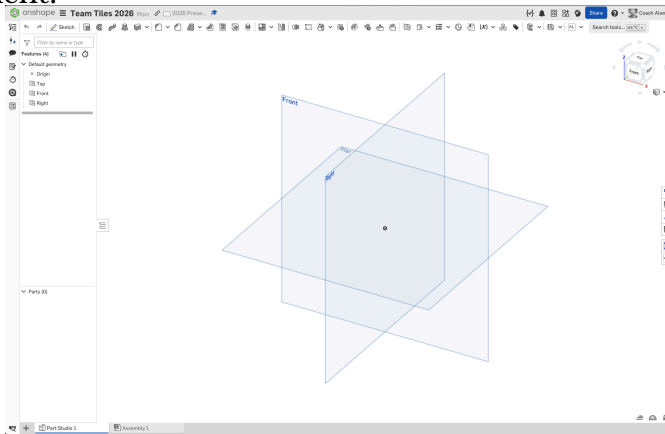


After we have done the folder, we’ll come back to this and Create a document. In this case, we are going to call it team tiles:

2.2. Create a “team tiles” document

The 'New document' dialog box in Onshape. It has a title bar 'New document' with a close button. Below it is a 'Document name' field containing 'Team Tiles 2026'. Underneath is a 'Document labels' field with the placeholder text 'Search labels'. The 'Document location' section shows a folder icon and the path '2026 Preseason'. Below this is a blue button that says 'No additional folders'. At the bottom right are 'Create' and 'Cancel' buttons.

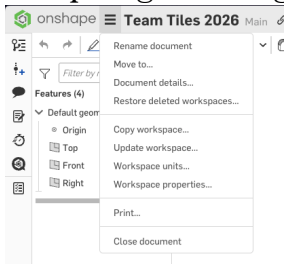
After we press create (blue button in lower right), it will take us to that document:



You'll see that this shows the name of the document in the upper right followed by what folder it is in.

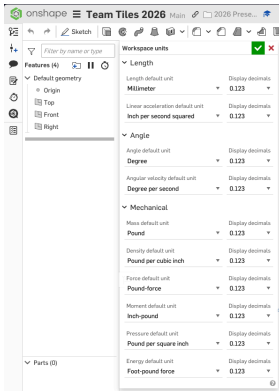
The next thing we will do is make sure the workspace units are set to millimeters for length. (If you are using GoBilda or REV parts, you will be much happier if you work by default in metric.)

To do this, click on the “hamburger” menu ($\equiv$) in the upper left after the Onshape logo. You'll get something like this:



2. For coaches...

Click on Workspace units and you'll get something like this:



You could argue (and I would agree with you) that you should change all of the units to be metric. But I typically only change the other defaults when I need to use them since I don't use them in many documents. (Personally, I change my account to have millimeter as the default length unit under preference)

You'll notice that at the bottom of the screen you see two tabs - Part Studio 1 and Assembly 1. These are the two you get by default. First, click on the "Part Studio 1" and rename it "Coach Tile".

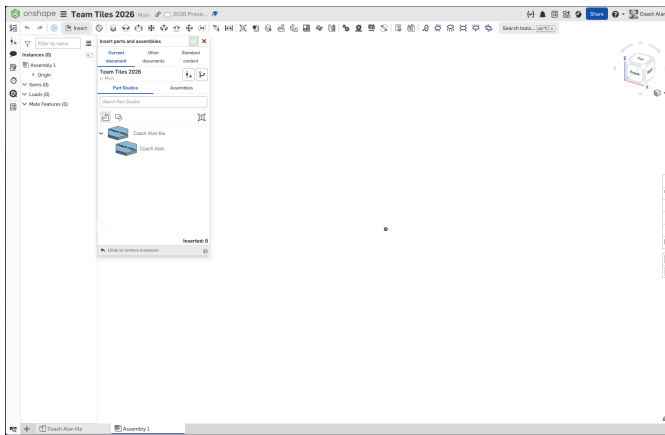
2.3. Create your individual tile

Since this is the same thing we'll have students do, go follow the steps for students in [section 3.1 on page 13](#) and then come back here for the next step.

2.4. Put it in the assembly

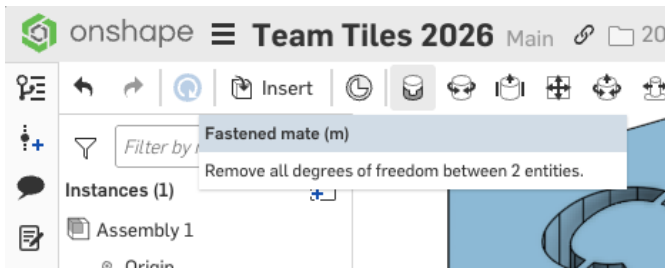
Switch to the tab called "Assembly 1". Now you want to select the word "Insert" on the toolbar (or press ) and you'll select the part you made in the part studio.

2.4. Put it in the assembly

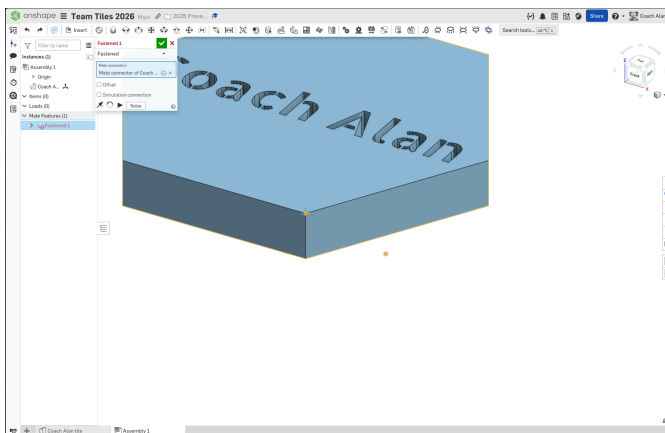


Yes, you could put multiple copies in but for now just put in one.

We'll select a fastened mate (this means that the two things you are connecting can't move at all with respect to each other.) . This is immediately to the right of Insert. (or you can press **m**)

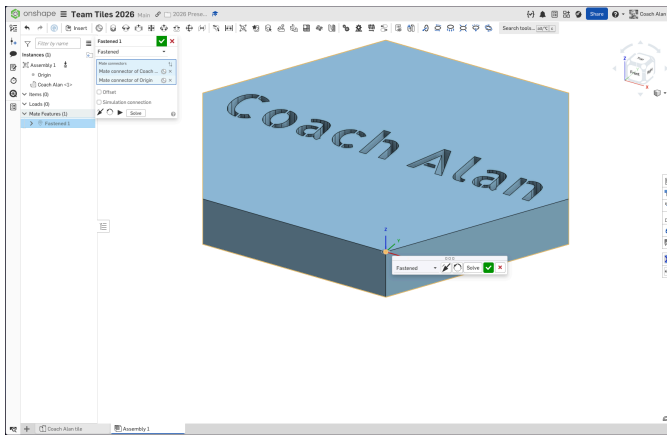


Then we'll select the lower right hand corner of our part (the top part) and the origin.



After we have selected both, we'll notice the part snaps to make that true.

2. For coaches...



Now we can press the green check. you'll see that there is no longer a symbol next to Coach Alan part saying that it has degrees of freedom.

Now we can close this (you don't have to save, like Google Docs it is always saved) and move on to our students making their own name tiles and attaching them.

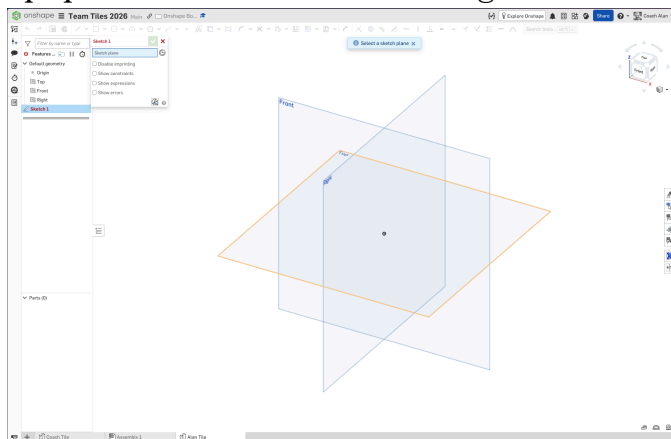
3. For students

If your coach hasn't done [chapter 2](#), stop and get them to do that. If you don't think that it is likely they will ever do it, then play paper rock scissors on your team and get the loser to go through [chapter 2](#)

Open the Team Tiles document for this season that your coach created. In the lower right, select the + and then Create Part Studio. It will be highlighted and will be "Part Studio 1" (if all of the other Part Studios have been renamed). Right click on it select rename, and name it "<your name> Tile" (where you replace <your name> with your name...) . While you don't have to rename the part studios, doing so will make it easier to find things later so it is a good habit to get into.

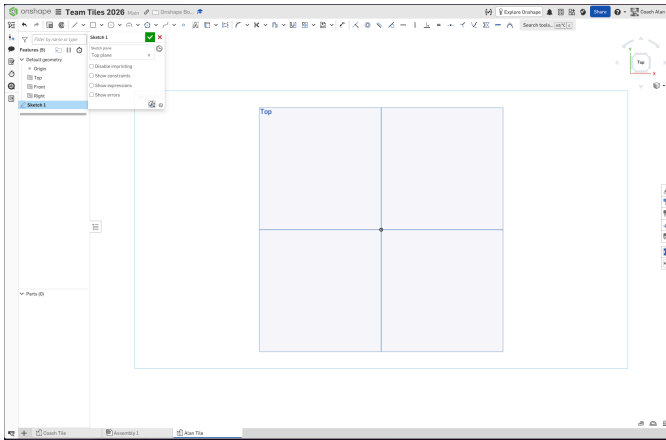
3.1. Making your team tile

Select "Sketch" up on the top (or press **Shift + S**). It will then ask you to select a sketch plane (this is the plane you are drawing on). In this case, let's select the "Top" plane so that we are sketching it as if we were looking down on it.

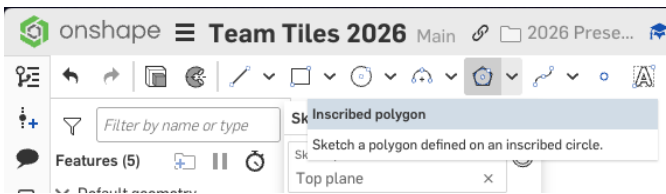


After this is done, press **N** on your keyboard. This will make it "normal" (perpendicular). I find that this makes it easier to work with as it now looks like I am working on a piece of paper instead of it being rotated in 3d space.

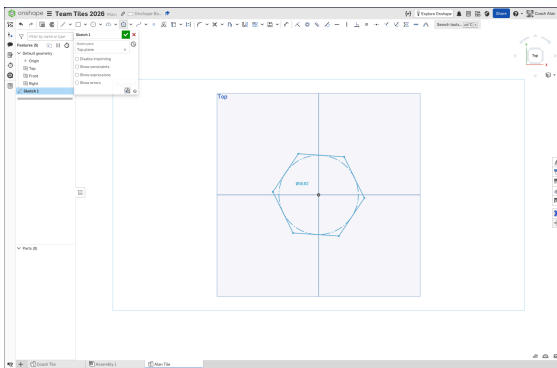
3. For students



The next thing we are going to is to select an inscribed polygon. (Or a polygon that is defined with a circle on the inside.) . We do that by clicking on the picture of the polygon in the toolbar.



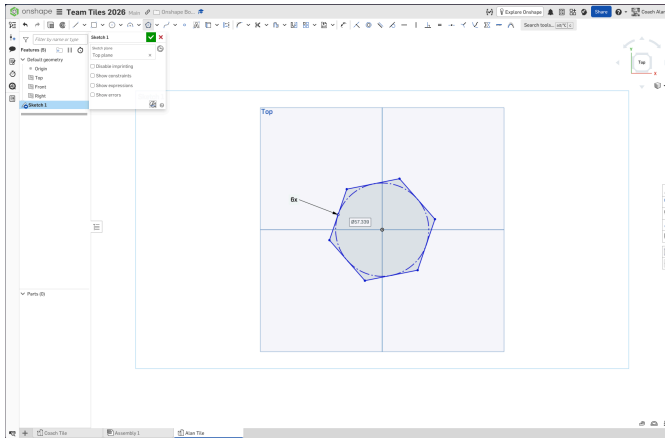
After we have clicked on this, we will click on the origin (the circle with a dot in it in the middle)



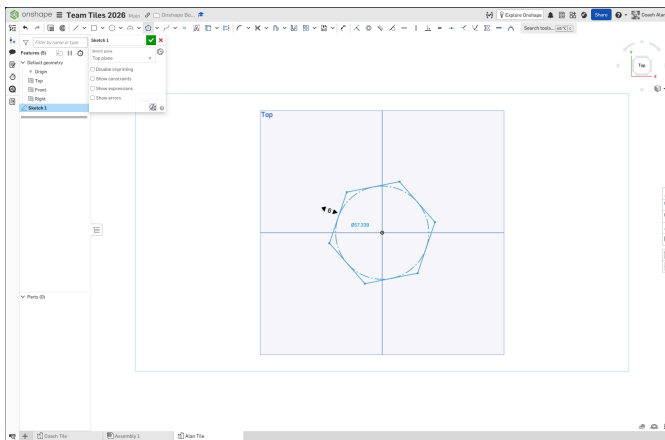
The next movement will change the size. It doesn't matter the size you make it because we will fix that in a minute.

After you click, the next thing it will ask is how many sides you want the polygon to be. The default is six and that is what we want, so we can just click to accept.

3.1. Making your team tile



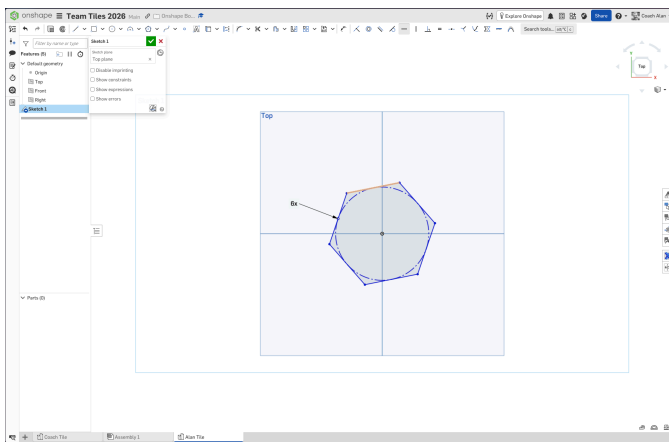
Then it will ask us to set the diameter of the circle. (I often mess up entering this here so I am going to show how to do it another way in a minute, but you can go ahead and enter in 70. This means that the circle (drawn with dashes and dots to indicate that it is for construction only) has a diameter of 70mm (or 7 cm).



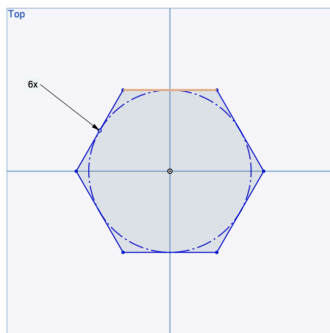
After you finish it, it will start making a second one. Since we only want one, press **ESC**.

The next thing we are going to do is apply a “constraint”. This means that we are saying something that must be true. For this, we are going to say that the top line must be horizontal. So we click on the horizontal line in the toolbar (or press **H**) and then on the top line of our hexagon.

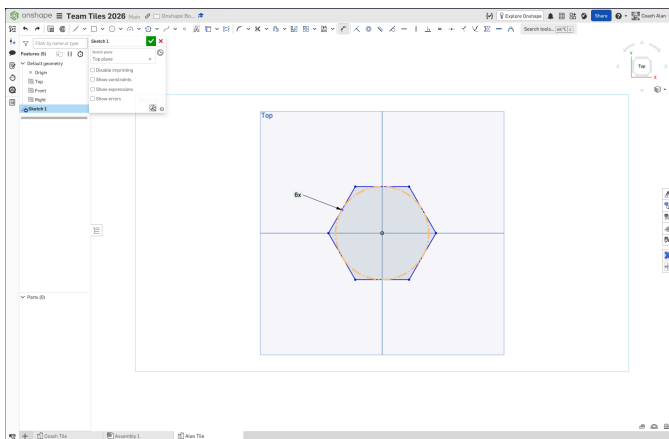
3. For students



You should see that your hexagon now looks like:

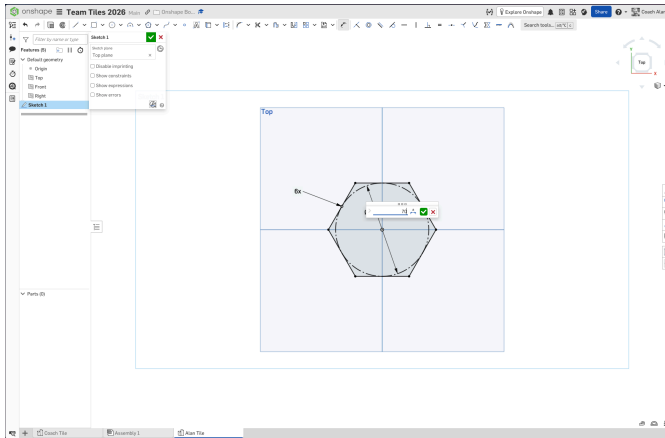


Now we are going to set the size of the circle. To do this, you can either select the dimension item or press **[D]** on your keyboard (for dimension) and then click on the circle.

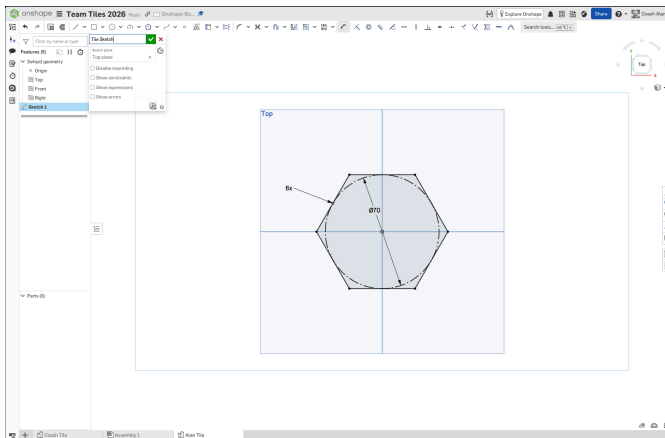


You can then enter in the dimension as 70.

3.1. Making your team tile



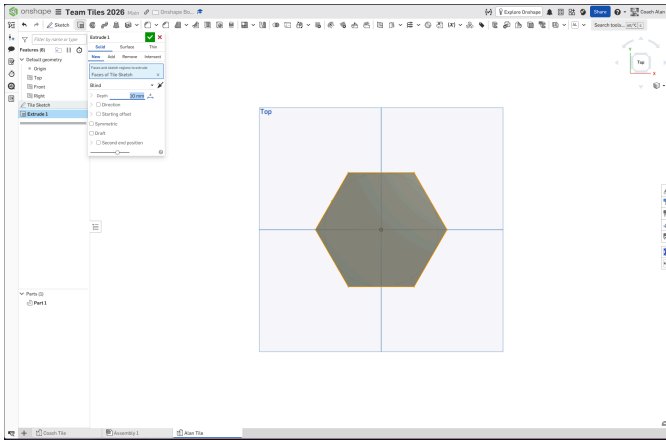
Then we can rename the sketch so we have some idea of what it is referring to and mark that the sketch is complete.



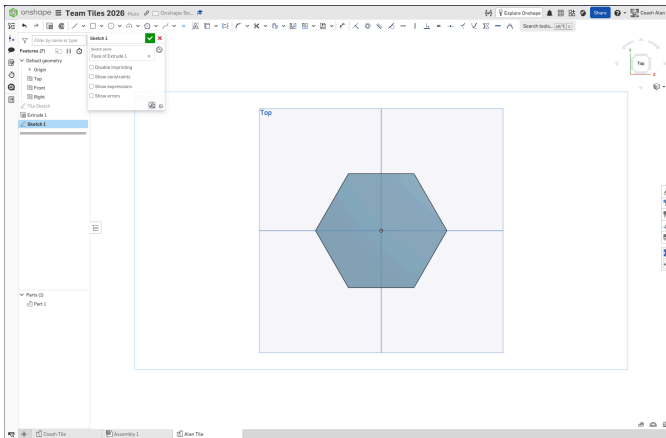
You can see here that the sketch is completely black. This is good because it means everything is defined and it can't change on you (more on that later). I renamed it to Tile Sketch and press the check button.

Now we are out of the sketch, and we will select extrude (the icon to the right of the word Sketch) in the toolbar (or press **Shift+E**). Extrude is like pushing a material through the sketch you just made and making it a certain size. Click on the sketch to say that is what you are extruding and make it a blind sketch of 10mm (1 cm). You'll see that now you have your first part down in the Parts list.

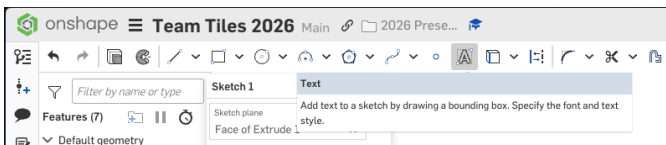
3. For students



Now we need to make another sketch to have our name on it. (Yes, you could have combined this with the first sketch) . So select sketch, and click on the top of your part.



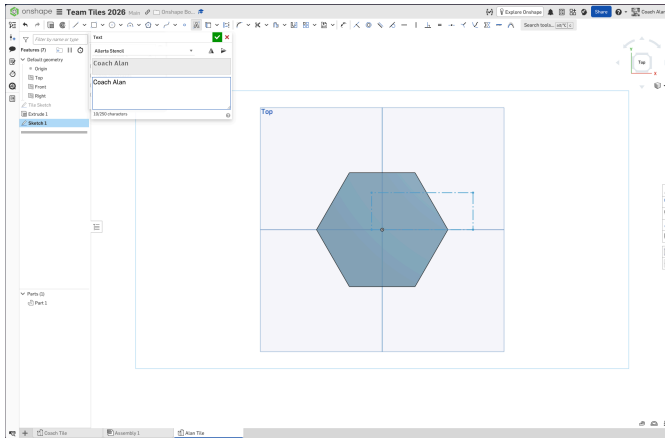
We now want to put text in our sketch. It is the icon that looks like the letter A in the toolbar.



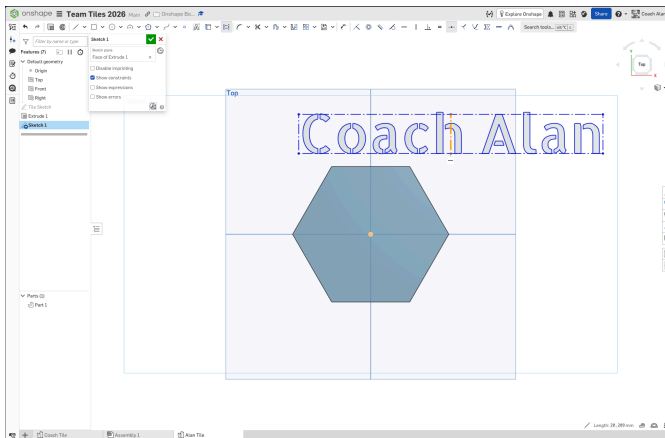
You then draw the bounding box. you can put it wherever and at whatever size because we are going to use constraints and dimensions to locate and size it.

After you have drawn the bounding box, type in the text and press the green check. (Obviously use your name instead of mine)

3.1. Making your team tile



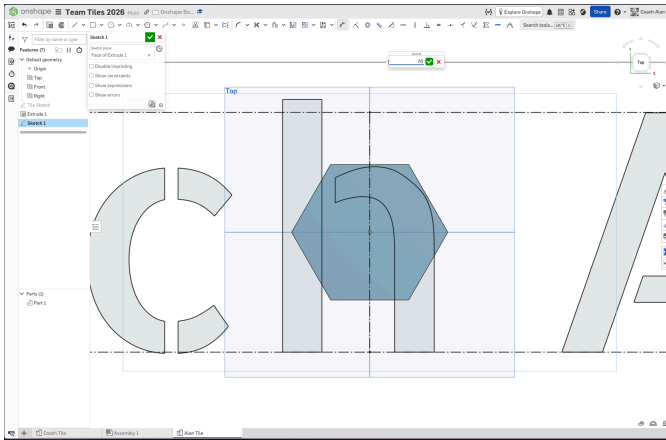
Now, click the midpoint constraint. (It is the one that looks like a line with a dot in the middle.) or you can press Shift+M. Click on the vertical construction line (construction lines are dashed lines) in the middle of the text and the origin.



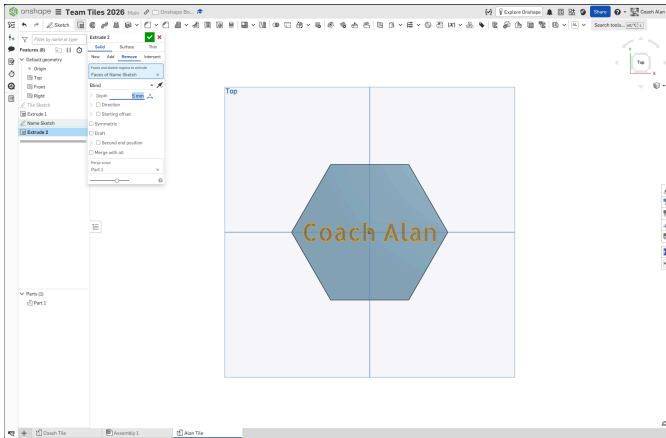
It will snap it to the origin.

Next, press d on your keyboard (for dimension) and the top line of the bounding box. Enter 70. (That is the radius of our circle, so we know it will fit.)

3. For students



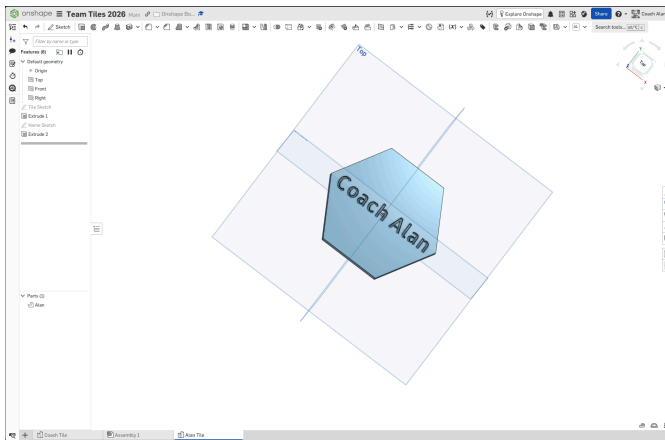
Now you can rename the sketch to be Name Sketch so you can find it. We extrude (**Shift+E**) just like we did before, but instead of “New”, we will select “Remove”. This will remove our sketch from the part we select. Notice we make the depth 5 so that it will just cut halfway through the part. Since we have a stencil font, it doesn’t matter. But if we had any other font it would change our singular part into lots of little parts.



We can click on Part 1 and rename it to be our name. The reason why I do so much renaming is it makes things simple later. Instead of wondering what Part 1 is, it is obvious.

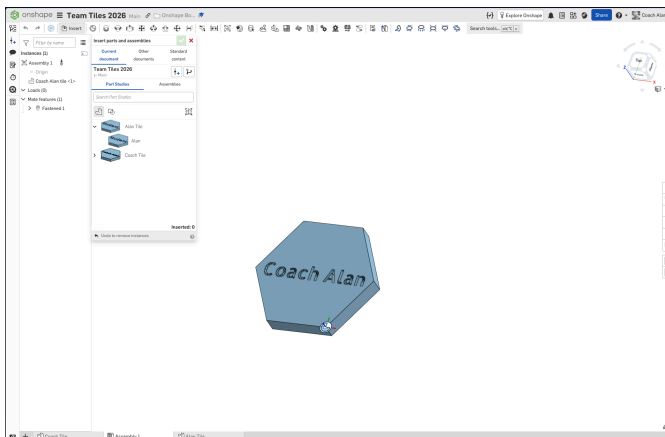
After you have renamed it, you can rotate by pressing the right mouse button and moving things around to see it from different angles. I am convinced that more time has been spent in CAD rotating items as a fidget than actual work being done.... (you can use the cube in the upper right hand to go to specific perspectives.) . You can also press **P** to show/hide the planes.

3.2. Connecting your team tile to the rest of your team



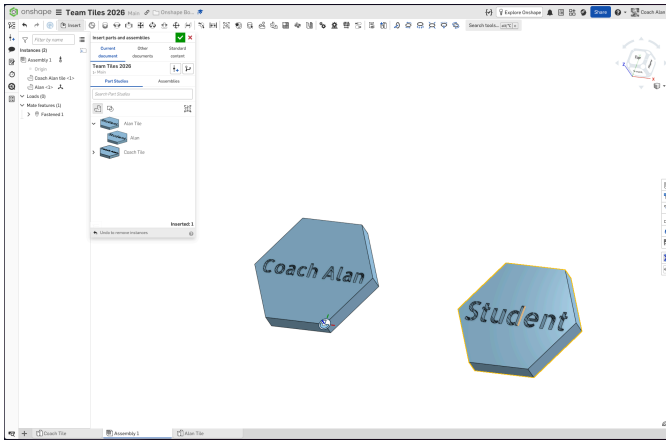
3.2. Connecting your team tile to the rest of your team

Go to the “Assembly 1” tab. Select Insert (or press ). Expand under your Tile Part Studio and you’ll see your part.



Select it and place it somewhere in the document that isn’t overlapping other things (just to make it easier to connect)

3. For students

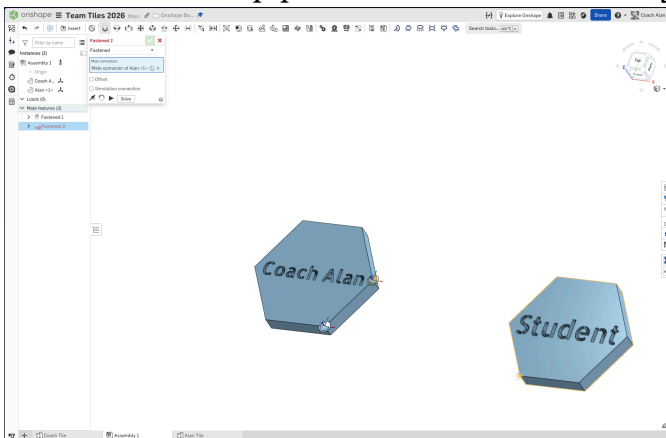


We'll accept this (by clicking on the green check mark) and then create a fastened mate (Press **M**)

3.2.1. Fastened Mates

A fastened mate says that two parts are connected with no degrees of freedom (ie, they can't move or rotate in any way) . This is almost like saying they are glued together. It is the most common mate that we will use.

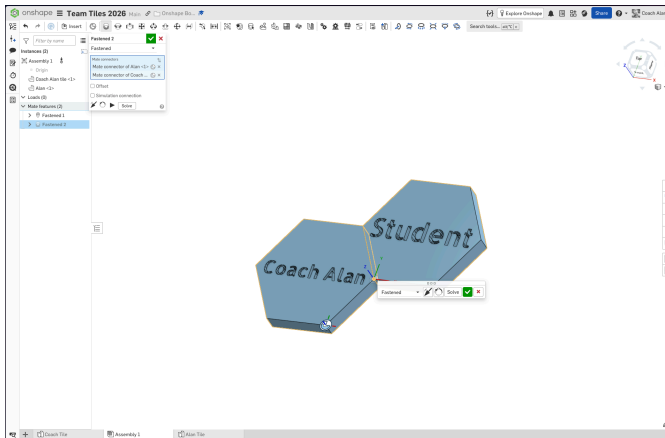
We need to select the point on each part where we want it fastened. Think about what two points you want touching. In the team tiles, you could pick the lower left corner of the “Student” tile and the right midpoint of the “Coach Alan” tile. For both of them, make sure the blue axis is pointing up. (This means you have selected the top part of the surface and they should both go together.)



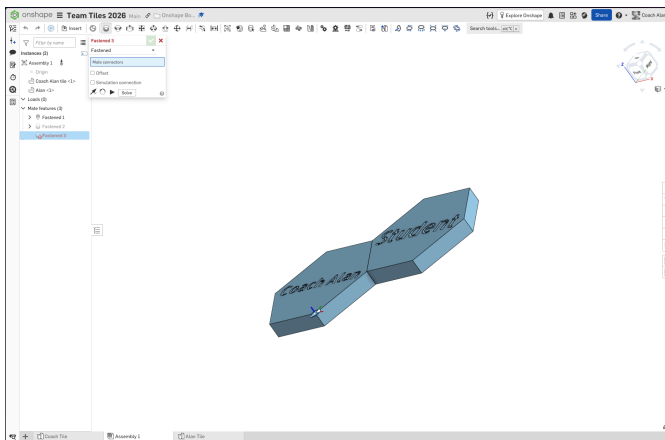
When you have the two points selected, you'll see the parts “snap” together. (This only shows the current mate. If you want to see what it looks like with

3.2. Connecting your team tile to the rest of your team

this mate and all others, then select “Solve”) . If it is what you want, press the green check mark.

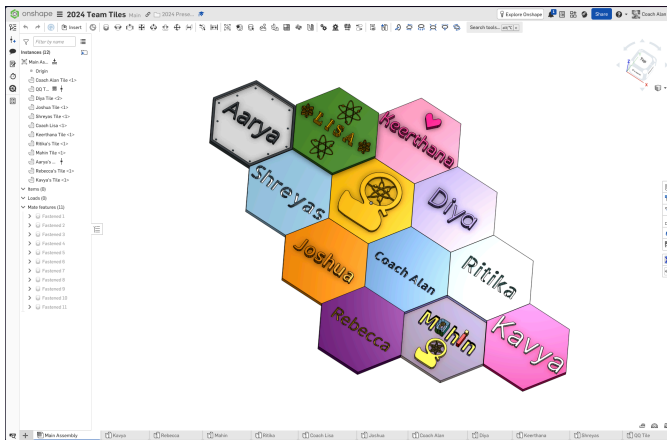


After it is fastened, you can rotate it around and it will stay connected.



As an example, here is a completed Team Tiles from one of the years that I coached. You can tell that many students decided to update their tiles to be a little fancy. One of the cool things is that if you update it on your part studio with just your tile that it will automatically update the assembly.

3. For students



While this was simple, I like starting each year with this because it makes sure that everyone has a working account and they can modify a team document.

Part II.

Modeling a simple Part

4. Spacers

We are going to start with the simplest useful thing that you can make in CAD and 3d print. This will help us go through the process with a simple thing that prints quickly so we can figure out where any problems are before we try more difficult things.

While you can buy spacers from GoBilda in a variety of sizes and shapes, you'll likely never have all of them and they are easy to design and print. A spacer is simply something that goes over a screw (or a shaft) that does not keep the screw (or shaft) from turning.

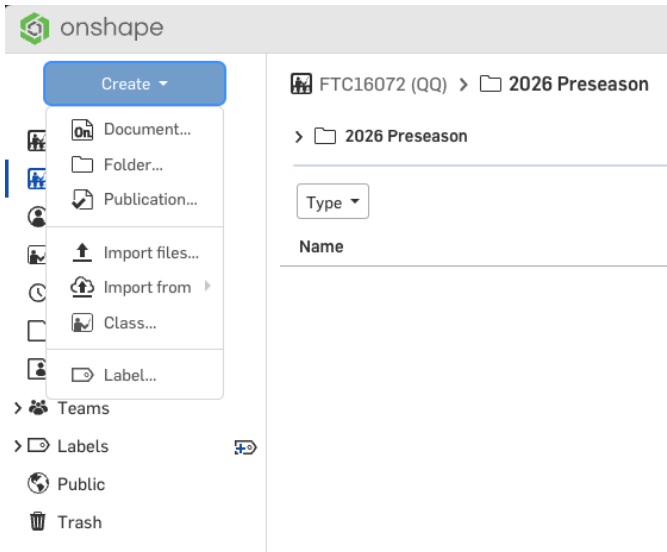
4.1. A simple spacer

Let's say that our screw is a little too long and we want to make sure that it doesn't stick out too far on the other side. GoBilda screws are M4 (which means they are 4mm in diameter). Let's say that we are trying to have the head 5mm from where it was.

4.1.1. Make in CAD

Let's go make a new document by selecting Create and Document. Call the document Spacers

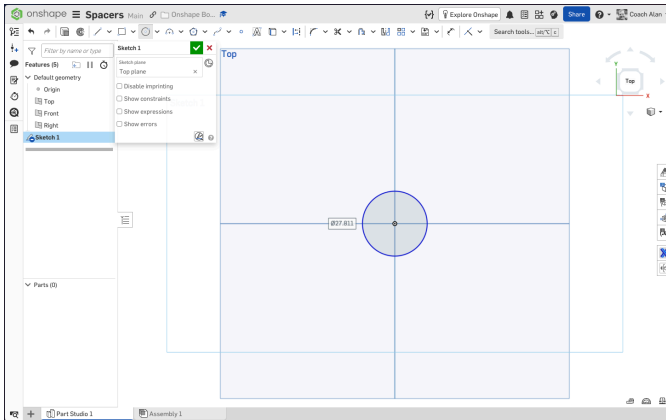
4. Spacers



Select “Sketch” up on the top (or press **Shift + S**). It will then ask you to select a sketch plane (this is the plane you are drawing on). In this case, let’s select the “Top” plane so that we are sketching it as if we were looking down on it.

After this is done, press **N** on your keyboard. This will make it “normal” (perpendicular). I find that this makes it easier to work with as it now looks like I am working on a piece of paper instead of it being rotated in 3d space.

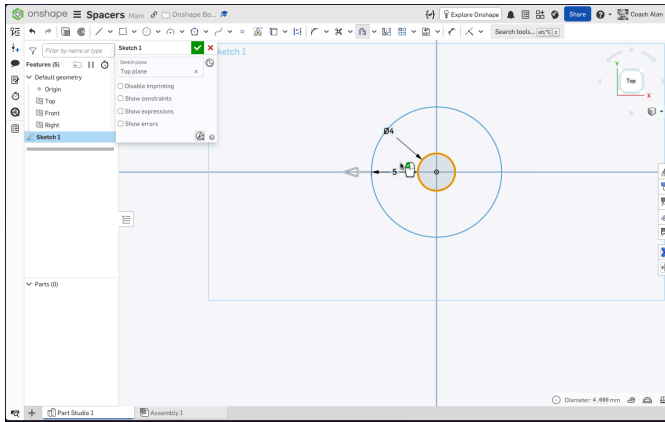
Select center point circle (or press **C**), then click on the origin, move it some distance and click (it doesn’t matter how far because we are about to set it.) . You’ll get something that looks like:



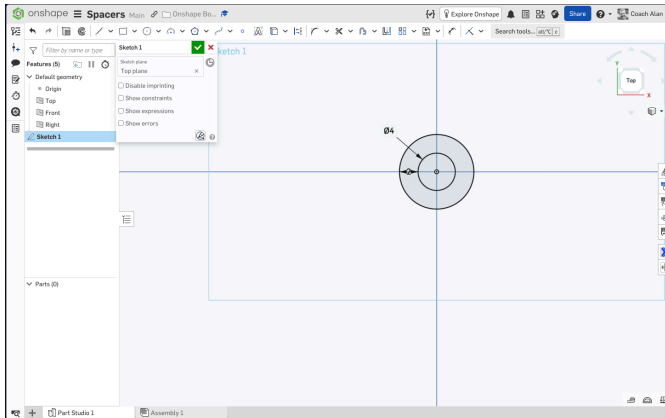
Type in “4” because that is what we want our inner diameter (often called ID) to be. If this is too small to see, you can press f to scale your window to fit just what you have (including the planes.)

4.1. A simple spacer

While we could just make a second circle of the size of our outer diameter (often called OD), we are going to use offset instead because it is a very useful thing to know. Either select Offset (or press **O**) and then select our circle.



Accept by pressing the left mouse button. It will then bring up a box for you to type in the offset. While normally I like a thickness of 3mm, that feels overkill for a spacer whose ID is only 4mm, so I put in 2mm here. Your sketch should now look like:

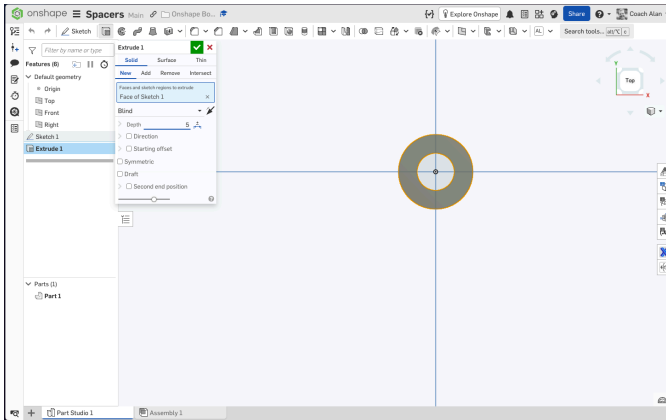


Make sure yours has black circles here and not blue ones. The black lines means it is fully defined. The inner circle is fully defined because we said it started at the origin and have given it a diameter of 4mm. (The symbol $\varnothing$ means diameter). The outer circle is fully defined because it is an offset of 2mm from the inner shape (which is fully defined)

One cool thing about using offset is that if you decide to change the inner diameter, the outer diameter will automatically change to have the same offset.

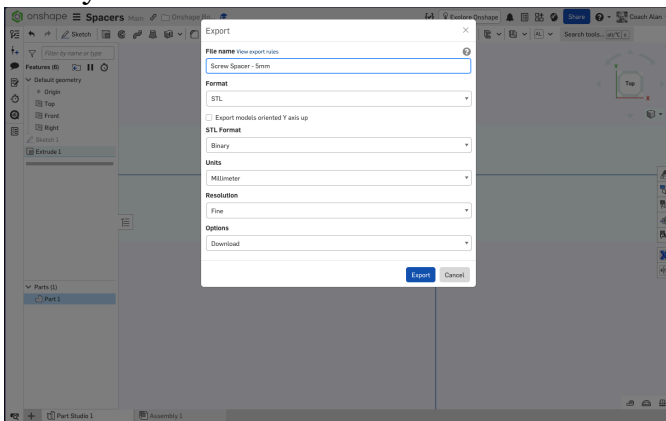
Now click on the section between the two circles and select Extrude (or press **Shift+E**) and make the depth 5mm and press the green checkmark.

4. Spacers



4.1.2. 3D printing our part

Right Click on “Part 1” and select Export... I change the File name here so that my computer isn’t littered with Part Studio 1 - Part 1 that I can’t tell what it is. In this case, I’ll rename to “Screw Spacer - 5mm”. It is important to know that this only affects the name of the file that is downloaded.

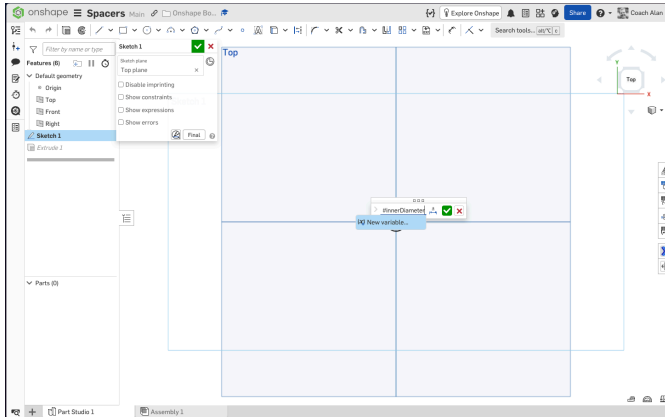


Press Export and it will download to the downloads directory on your computer. You can then bring it into your slicer. (This is 3d printer dependent). Make sure the bottom surface is flat on the bed so that it will print vertically.

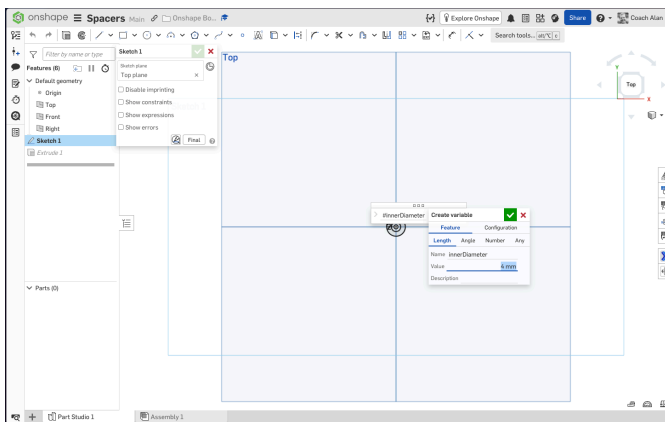
4.2. Using variables

While this is pretty cool, it is going to get old doing this over and over again for every spacer that we need. It turns out that we can do better. Let’s right click on “Sketch 1” (we can rename sketches and if we have multiple ones we should)

and select “Edit”. Click on where it says $\varnothing 4$ and type in “#innerDiameter”. The # character means use a variable. (We can either choose from existing variables or we can create new ones here. But since we don’t have any yet, we will have to create a new one.) . Either select the green checkmark or Press **Enter**.



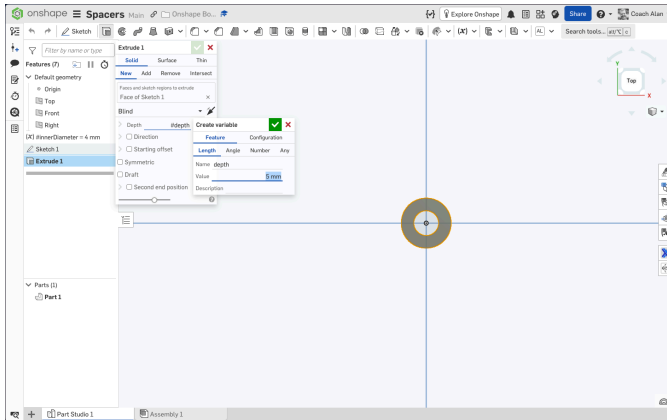
It will then ask us to create a variable. (We will have it as a length and a feature. You can see about Configurations in [Appendix C](#).) . Variables in Onshape have units attached to them and you don’t have to use the default units of the document.



When we press all of the green check marks, we’ll see that there is a gray fx next to the diameter showing that it is calculated. Click the green check on the sketch.

You can double click on “Extrude 1” and change to #depth. That will have us create the variable.

4. Spacers



Now you'll see two variables in the "feature list". They are evaluated from top to bottom, so your variable definition has to be before it is used. To change the value of a variable, you can double click it. So if you wanted to make a spacer that went over a 5mm D shaft for 10mm you would just change the innerDiameter to be 5mm and the depth to be 10mm and you can see the whole thing regenerate.

5. Simple Robot sign

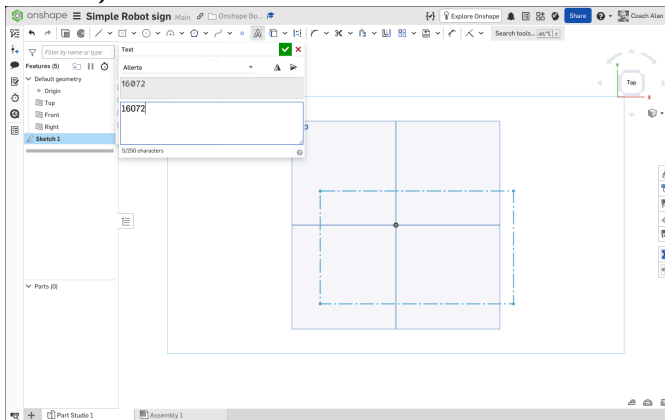
This may be obvious, but before you begin in CAD figure out what the problem is you are trying to solve. This is helpful to write down because it can be clarifying for what is important.

For our next part, we are going to make robot signs. From the competition manual (R401) we know that it has to be minimally 6.5 inches wide and 2.5 inches tall and the numeric part must be at 2.25 inches tall (+/- 0.5 inches). We need 2 red signs and 2 blue signs. So let's assume we have 3 different color filaments and we are going to make this with our 3D printer. To make this simple, we are going to attach it with velcro on the back. That way we can easily swap out the parts.

So let's create a new document called "Simple Robot sign" (if you don't remember how, reread [chapter 4](#))

We will create a sketch on the top plane (again from [chapter 4](#)) and press **N** to make it so we are looking at it top down.

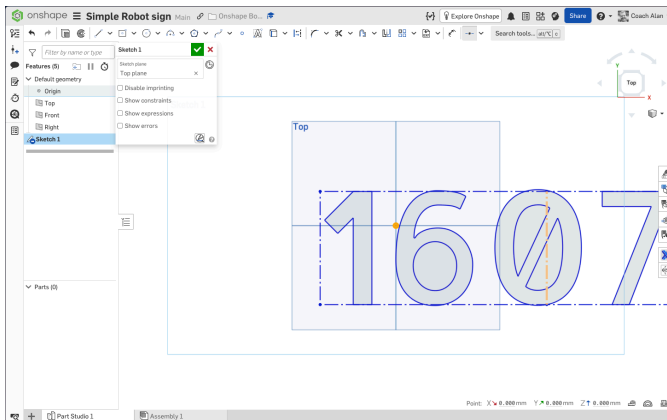
First select Text and make the blue rectangle in the middle. (don't worry about size, we'll adjust later). Put in your team number. (The team I coach is #16072¹)



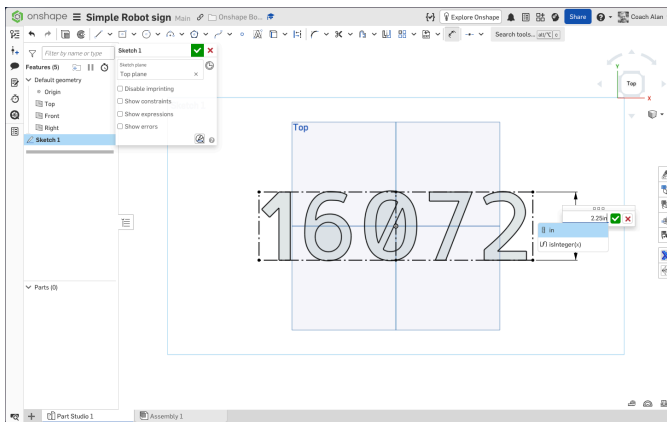
Now select Midpoint (or press **Shift+M**). Select the origin and then select the vertical construction line (construction lines are there just to help with construction and they are dashed) in the middle of the text.

¹Go Quantum Quacks!!

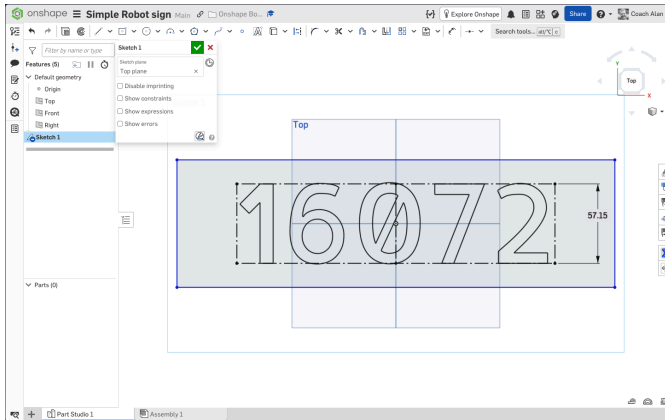
5. Simple Robot sign



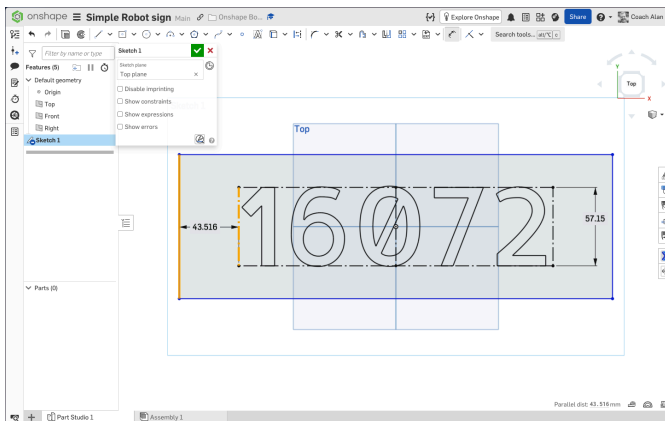
Let's give a dimension to the number since we know the height has to be at least 2.25 inches. Select Dimension (or press **D**) and click the right side and type in 2.25in.



Now add another rectangle (select center point rectangle or press **R**). Don't worry about its size. Just make it bigger than the other rectangle. When it is drawn after clicking you can press escape to not need to enter in dimensions.



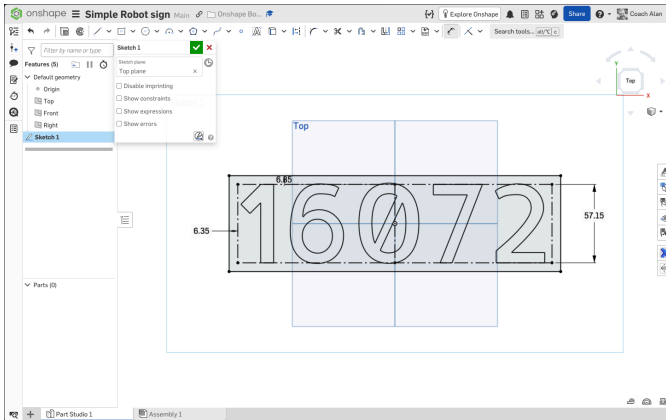
Now let's give a dimension between the left side and the text of .25 inches. (minimum spacing from the competition manual) . First we select Dimension (or press **D**) then we click on the left construction line of the text and the left line of the outer rectangle.



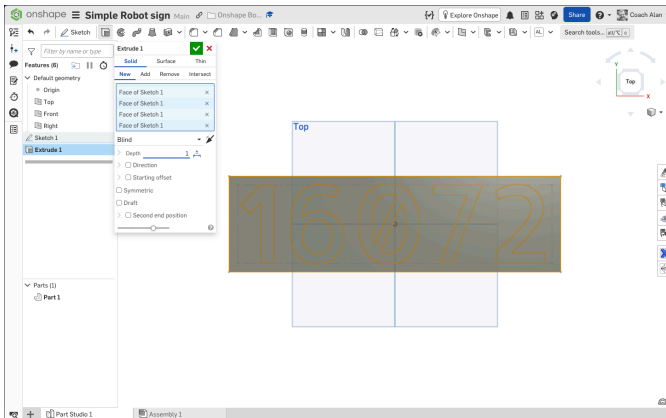
Now set this to be 0.25 inches. (You'll notice that it shows it as 6.35mm)

Now do the same thing between the top line of the text and the top line of the rectangle. Your sketch should now look like:

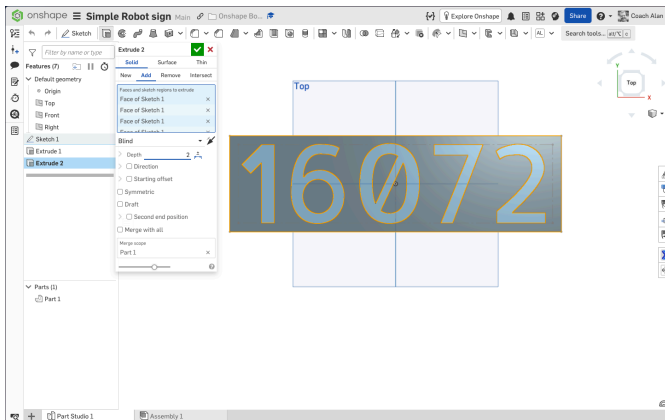
5. Simple Robot sign



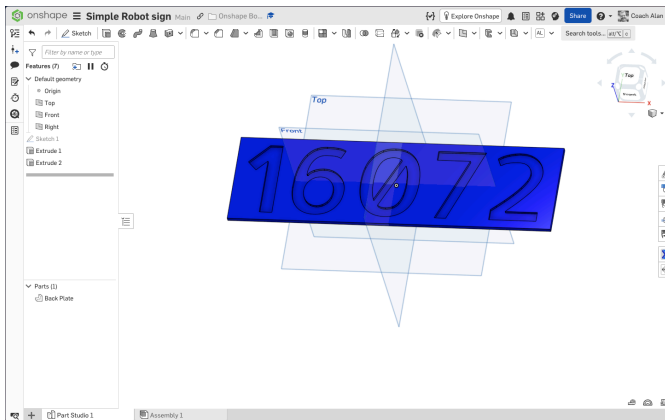
Now that we have our sketch, we'll go to making our parts. Select Extrude (or press **Shift+E**) and select everything so you have just a rectangle. Extrude it with a depth of 1mm.



Now, go in and edit the sketch. Select Extrude (or press **Shift+E**) and select everything except the numbers. Extrude it (adding it to the existing part) with a depth of 3mm. (This will make the gap for the numbers 2mm because the first mm is in common with the back plate.)

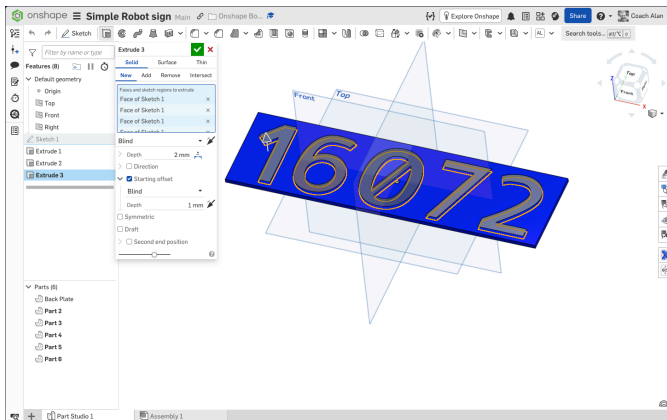


Now let's rename the Part 1 to be "Back Plate" and change the appearance (right click on the Part and Edit Appearance and select the blue color.)

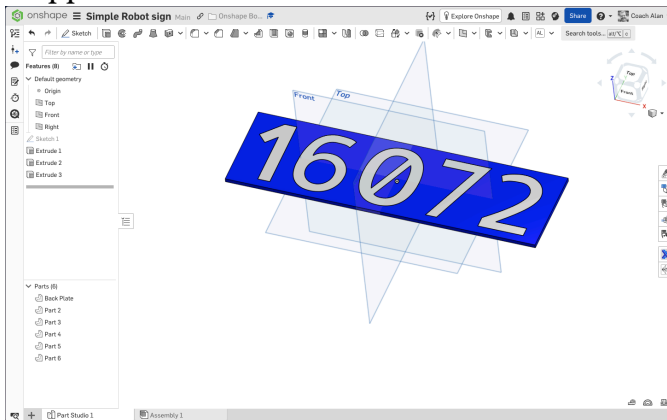


Now let's go back to the sketch and select all the number parts and extrude them (**Shift+E**) 2mm (so they'll fit in) . We also do a starting offset of 1mm so that they'll look correct in here.

5. Simple Robot sign



Now let's select part 2, part 3, part 4, part 5, part 6 and right click and set the appearance to be white.



Now you need to export each part like we did before in [subsection 4.1.2](#) Make sure you print 2 back plates in blue, 2 in red, and 4 of each number. Now glue the numbers inside of the gap. Put velcro on the back of the plate and then on your robot where the plate should go.

Friendly tip - My team likes to point a color sensor at the back of the number plate so the robot knows whether it should run the red autonomous or blue autonomous. You're welcome....

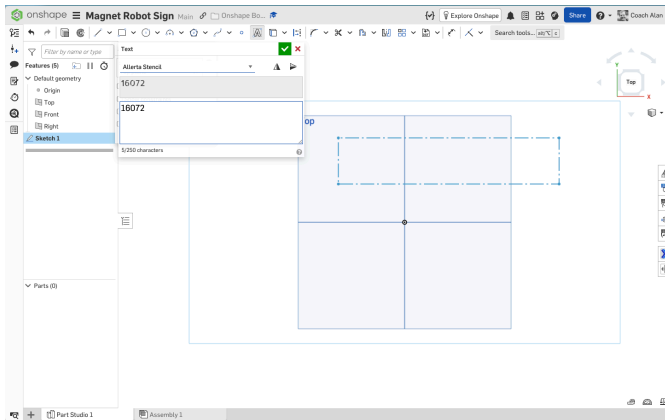
6. Magnet Robot sign

Ok, so we ended up with robot signs and we can play the game. But what if we wanted to make robot signs that screwed on and had magnets in them so that we just changed out the background color.

To do this, we start out by thinking about the design. What we want is 2 pieces. The first (backstop) that is all white and has a place that can screw in and ideally the holes are flush with the surface so the background. ¹ The second is the background color that needs a place for magnets.

Let's start by making a new document called Magnet Robot Sign, and make a sketch on the top plane. Press **N** so you are looking at it like a piece of paper. (These are all from [chapter 4](#))

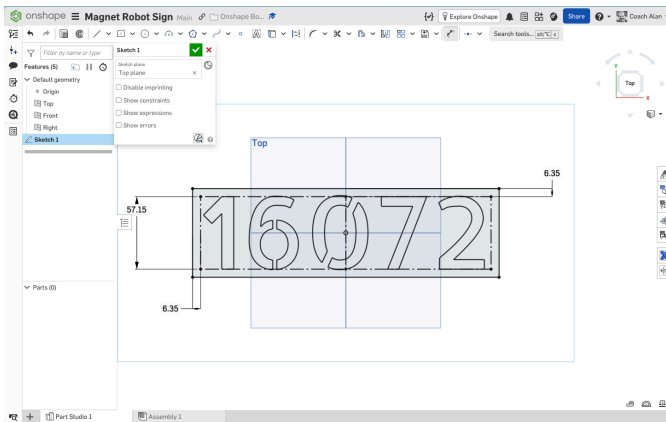
We are going to insert text (don't worry about where or the size, we'll set that later.) . This time make sure the font you choose is Allerta Stencil. A stencil font has no enclosed parts. We want this so that we can have only a singular background to lift up.



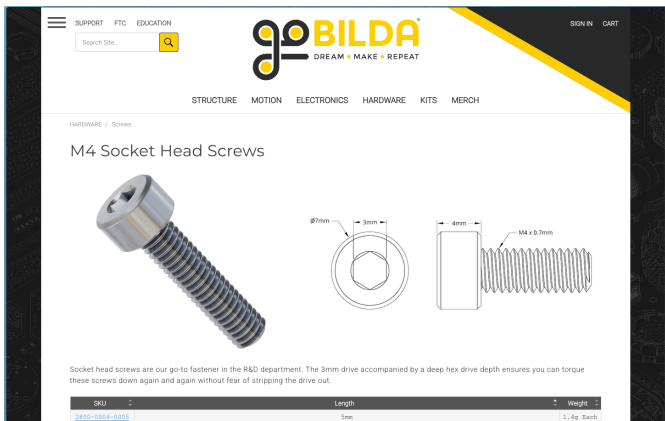
Follow the same steps from [chapter 5](#) to get the rectangle surround it with the same border. You should now have something that looks like:

¹Those of you that are familiar with woodworking might be thinking of countersinking . This is similar but since the screw heads are cylinders instead of angles, we call them counterboring

6. Magnet Robot sign

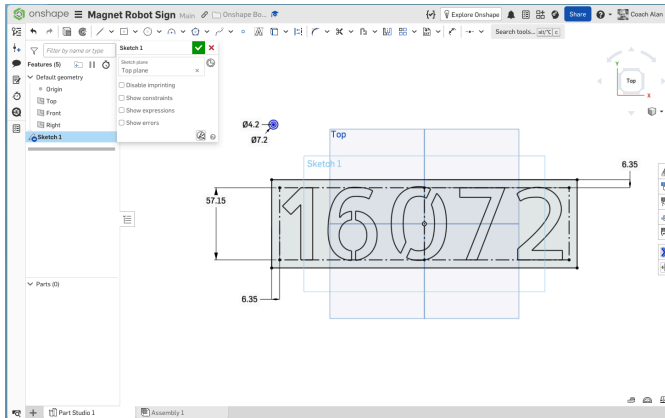


(I did move the measurements to make them easier to read. To do that, just click on them and drag them to where you want them.) . Now in this case, we want the bottom to be thick enough to hide the screw head. We go look at the data sheet for the screws we are using:

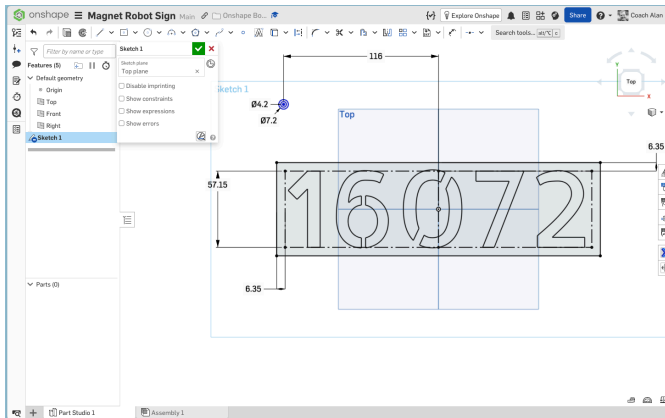


You'll notice that the OD (outer diameter) of the screw is 7mm so we need a hole that size for the head to rest in. The screw part is 4mm. Because 3d printers are not exact, we like to add a little extra to both of these so we have space. Typically .2mm is plenty.

So let's create two center point circles that share the same center point where the outer one is 7.2mm and the inner one is 4.2mm. Don't worry about exactly where to put it for now. Just make sure you stay clear of your sketch and the axis so that Onshape won't try to "help" you by creating additional constraints.



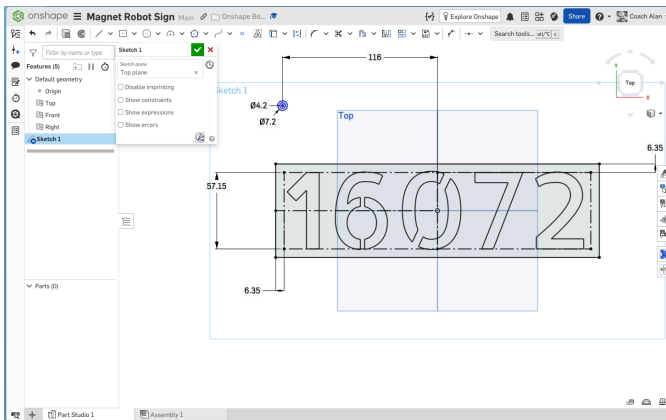
Now we are going to connect this to our robot, so we'll be happier if we do it on an 8mm pattern (GoBilda is on an 8mm pattern²). To do this, let's first do a dimension from the center point to the midline and make sure it is a multiple of 4. (since we will be doing it symmetrically 4 is $\frac{1}{2}$ of 8). We did 116 because that is close.



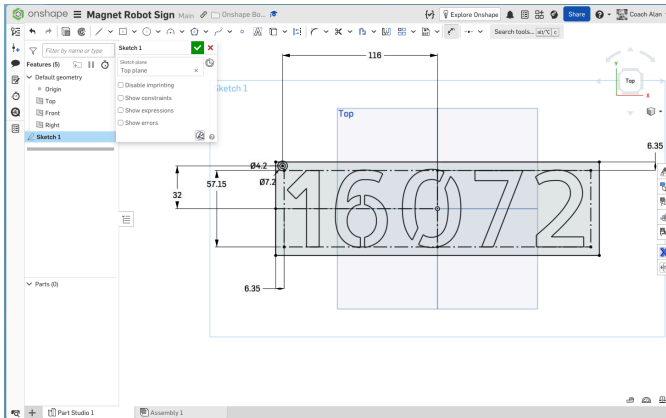
Now let's create a construction line (a construction line is a line that is just to help us making our sketch but is not part of our part. It is shown in dashes.) . To do this, select construction or press **[Q]**. Then select the line or press **[L]**. We are going to draw a line from the origin left to the outer box. Because you are connecting it to the outer box, it knows its length, because it is from the origin, it knows its starting position and Onshape guessed we wanted a horizontal line because we drew along the axis so it went ahead and gave us that constraint. It should look like this:

²So is LEGO, by the way....

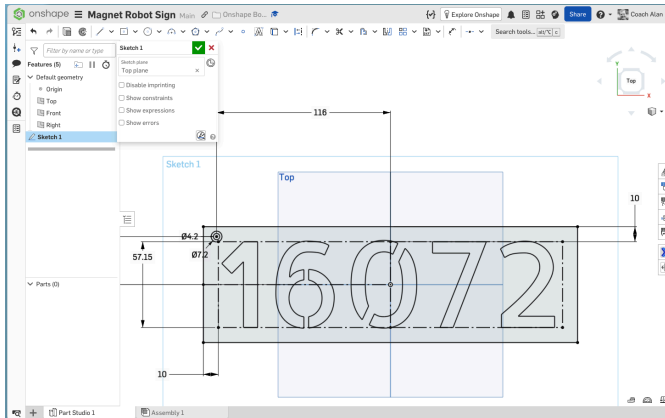
6. Magnet Robot sign



Now we do a dimension of it from the center line. We'll do 32mm. You should get something like this:

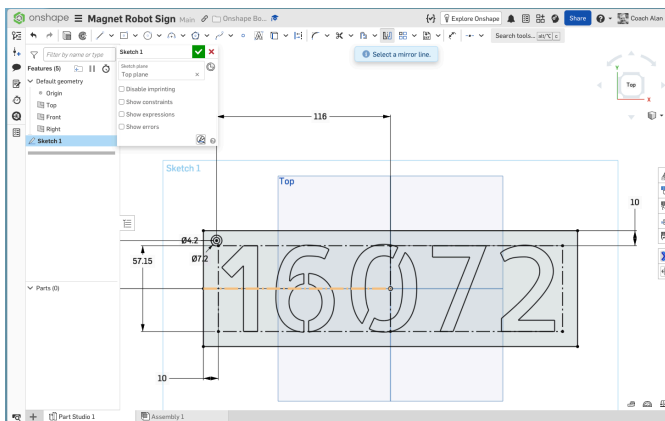


Hmm. There is a problem. Our hole goes over the edge of our sign. No big deal, let's just make the border bigger. Since we really want at least 2mm of material, let's change our borders from 6.35mm (or .25in) to 10mm.



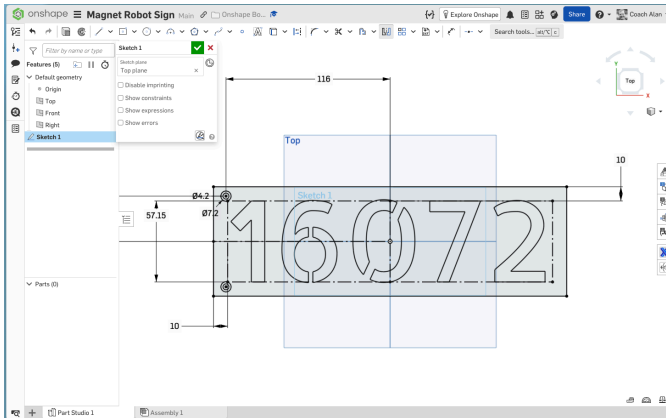
But we want four holes, and we only have one. This allows me to show one of my favorite tools. Click on the Mirror tool (looks like an “L”, a construction line, and a symmetric “L”).

The first thing it wants is for us to select a mirror line. This is the line we will mirror around. Choose the horizontal construction line we made.

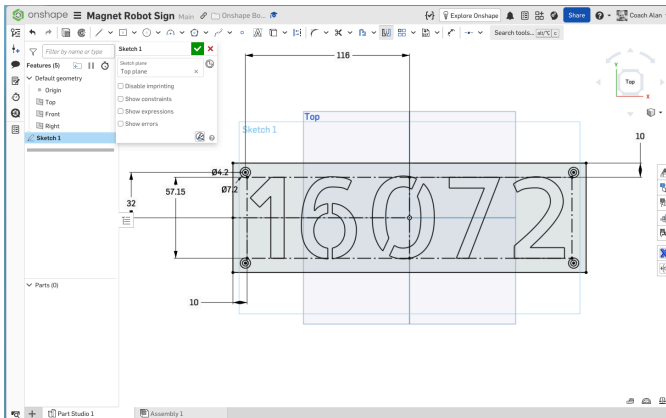


Then it will say to select the entities to mirror. Click on both the inner and outer circle. This will give you two.

6. Magnet Robot sign

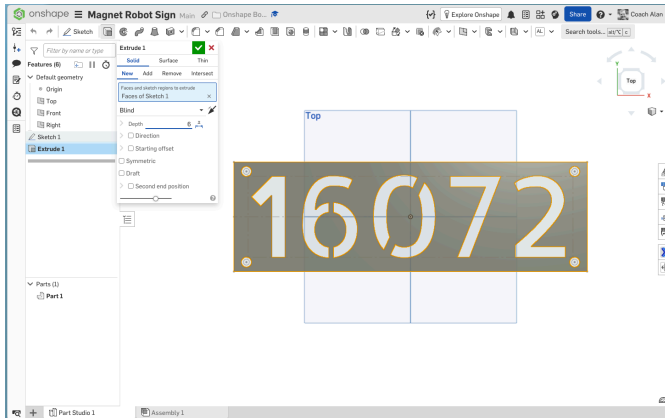


Now we do the same thing again, but this time select the vertical construction line and select all four circles. You should now get something like this:

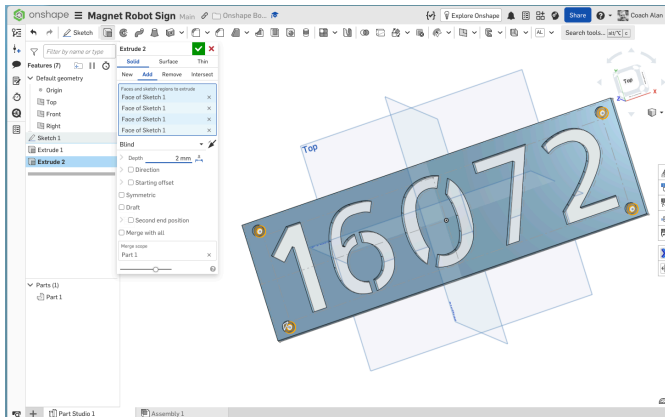


One of the really cool things is that if you change anything with the hole in the upper left (diameter, position, etc) then it will change it on all four. Lets say that you decide you want your screw hole to be 4.1mm instead, just change it in one place!!

Ok, so now lets extrude (**Shift+E**) the entire plate except the holes by 6mm.

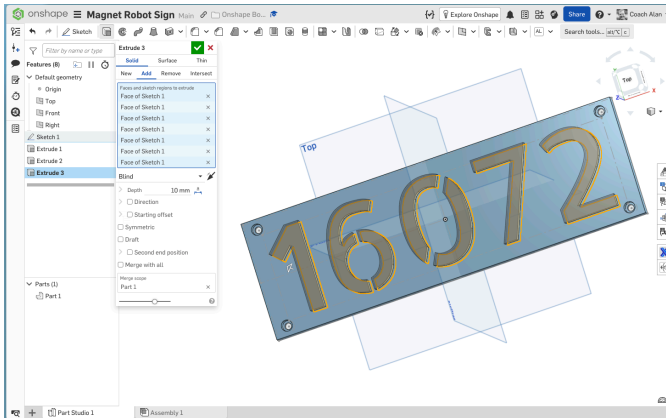


Now we'll add in the part we want for inside of the holes. Make the sketch visible (by clicking on the eye with the slash through it next to Sketch 1) and then select the outer ring of the hole. We'll extrude it by 2mm and select Add so it will make it part of Part 1. (Hint: Zooming in sometimes makes it easier to select small portions.)

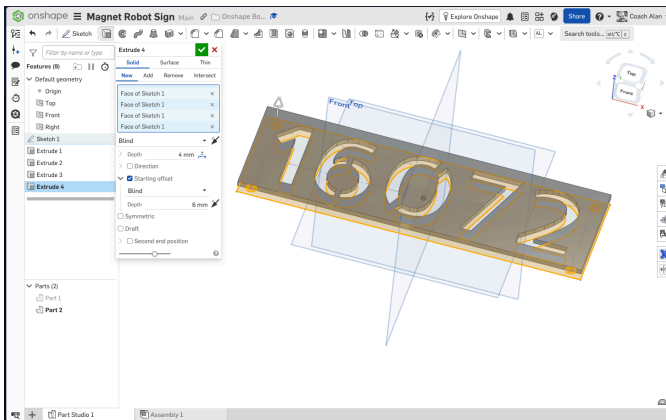


Now we'll select the numbers. Since we want a place for magnets (the magnets we'll use are 5mm diameter and 2mm height) then that means we need at least 4mm. Since the plate is already 6mm, let's extrude the numbers by 10mm.

6. Magnet Robot sign

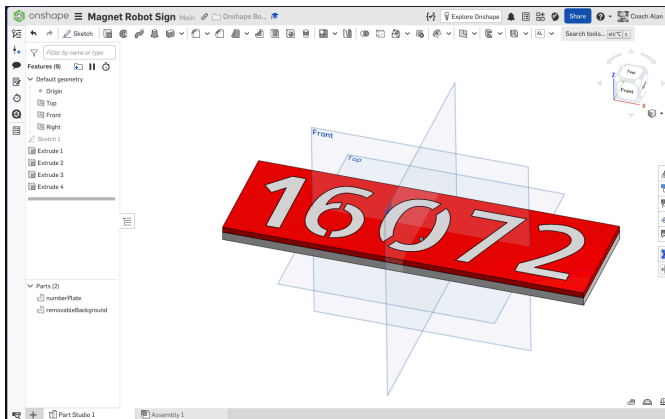


The easiest way to select the appropriate parts from the sketch is to hide “Part 1”. Move your mouse over it and click the “eye” which will move it to an eye with a slash through it. Select the holes and the plate (but not the number) and extrude it for 4mm but with an offset of 6mm. (This will make the two parts look like they are correctly positioned when we show both)



Now, let's rename Part 2 to be removableBackground. (right click on Part 2 to get to Rename). Right click on it again and select edit appearance and choose red.

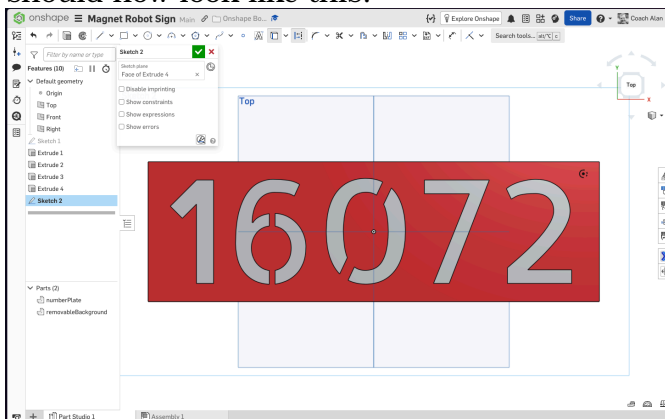
Now, we'll rename Part 1 to be numberPlate and change its appearance to be white. Make it visible and Hide Sketch 1 and you should get something like this:



Now we need to make the holes for the magnets. Since the screws are made out of steel magnets will stick to them, so let's have the magnets over the holes.

Start by creating a sketch but instead of clicking on "Top Plane" click on the top surface of the removableBackground. (Remember to press **N** so we can work on it top down)

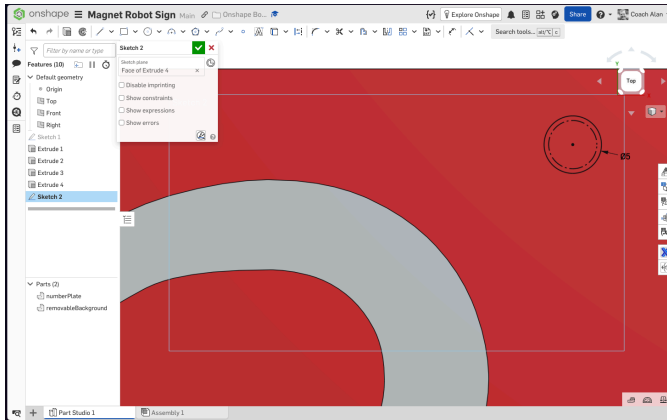
The best way to reference something else is to use "Use". You can either select it from the top or press **U**. Press **N** again and you are now looking at the back of the number plate. Select the construction line (press **Q**) and then click on the upper left hand hole. Press **N** again and you'll go back to seeing it from the top. You'll notice that the top left hole from the back side was the top right from the front side. This is fine, we'll just mirror opposite of how we did before. It should now look like this:



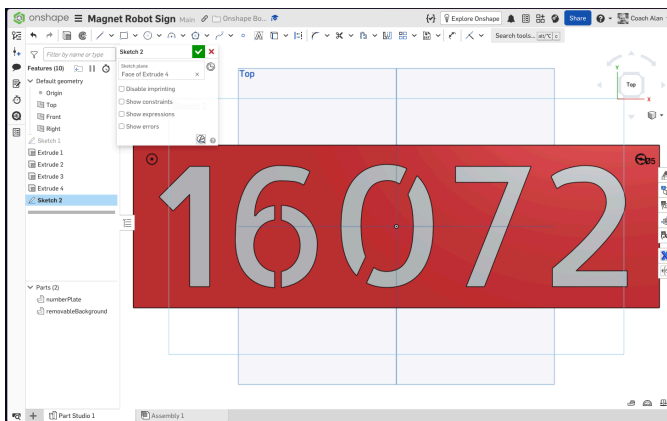
The first thing we need to do is create a 5mm circle. select the center point circle (press **C**) and click on the center dot. Then give it a size of 5mm. If it draws it dashed then select it and press **Q** to change it from a construction line to solid. This is one of those where it will likely be easier if you zoom in.

6. Magnet Robot sign

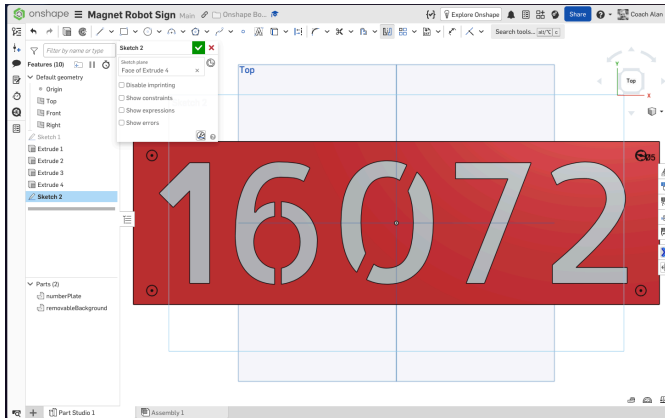
Here is what you should have.



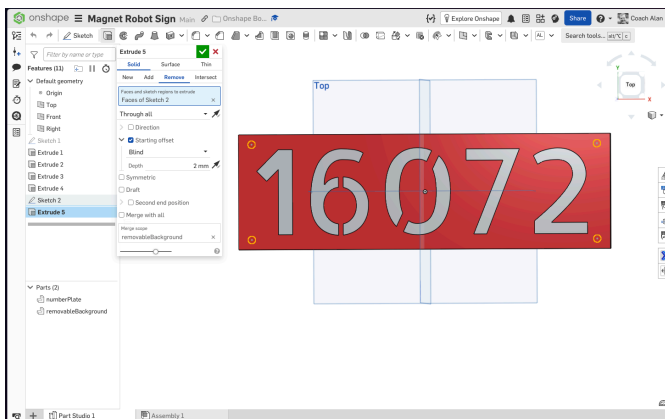
You can press **F** to zoom to fit. It turns out we can use the axis to mirror (we don't have to draw a construction line). So click on mirror, select the vertical axis and select the hole. (There is no need to mirror the construction circle)



Now we'll mirror the top two holes onto the bottom. (Select Mirror, choose the horizontal axis and select the two circles.)

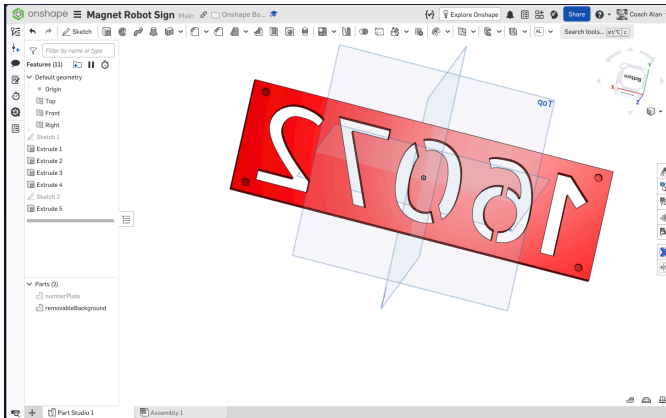


Select Extrude (**Shift+E**) and make sure the “Faces of Sketch 2” are selected. We’ll do a remove and this time instead of “blind” we’ll do a “Through All”. That way if we change the thickness of this part later we won’t have to change our extrude. We’ll do a starting offset of 2mm (so that the magnet isn’t visible) and select a merge scope of the “removableBackground” so the extrude knows which part we are removing from. If we did numberPlate here as well it would also make the hole go through it which isn’t what we want. Make sure the arrows here are going the correct direction.

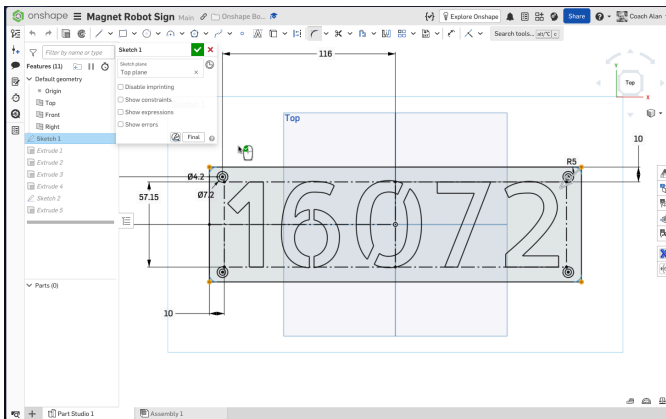


If you hide the numberPlate and turn it around you’ll see the places for our magnets.

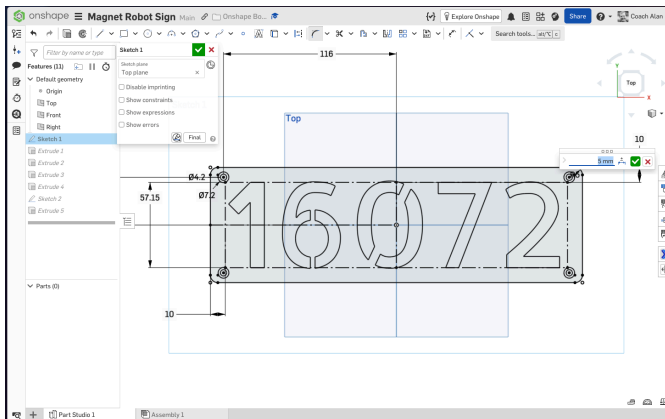
6. Magnet Robot sign



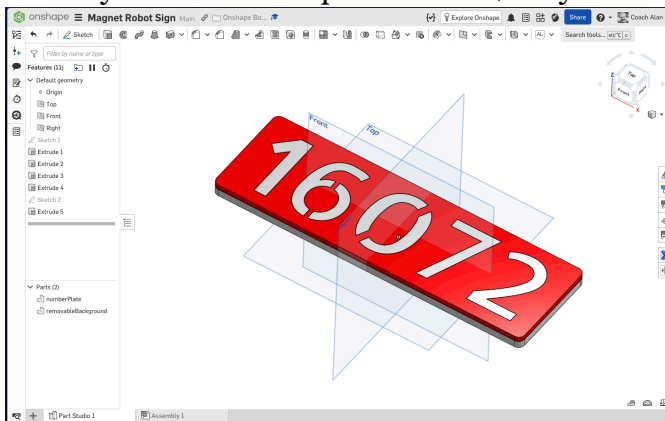
One of the cool things about everything based off of a sketch is that we can just make a change there and it affects everything. Right click on Sketch 1 and select Edit. Hit **N** twice so that it looks like we are seeing it from top down. Now select Fillet (**Shift+F**) and click on the four corners.



Right click on an empty space and it will ask you to set the radius of the fillet. The default of 5mm works well here so just accept it by pressing the green check.



Accept your sketch (Green check mark) and you'll see that it affected your parts. If you make both parts visible, they'll look like this:



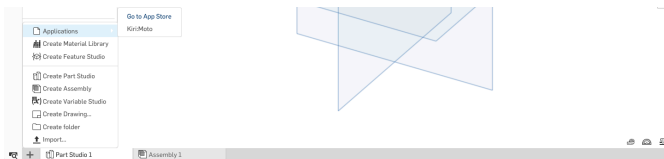
Not only does fillet look cool, it also makes your parts less likely to break because they don't have sharp edges.

Now you need to export each part like we did before in [subsection 4.1.2](#) Make sure you print 2 back plates in white, 2 removableBackgrounds in red, and 2 in blue. Now glue the magnets into the holes on the removableBackgrounds (you'll need 16 magnets in total. You can easily get 5mm x 2mm magnets on Amazon)

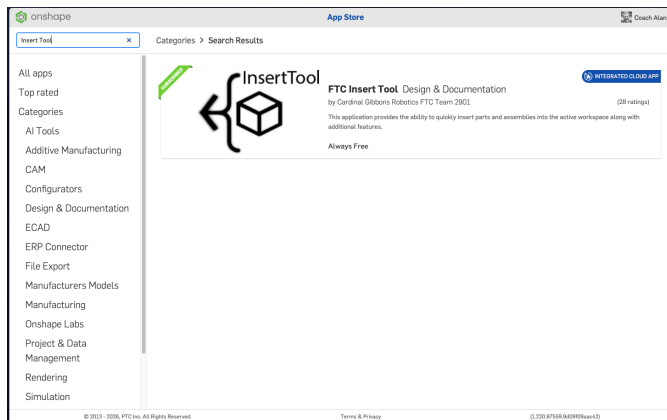
7. Cover end of U-Channel

Several years there are game elements on the field that could get stuck in your robot. (The samples in Into the Deep (2024-2025) are a great example.) . Since there was a possession limit, getting these pieces stuck was “no good”. So let’s create something that just seals the empty front of our U-channel and is slanted so that pieces get pushed to the side (like a cow catcher on a train).

Create a new document called “U-Channel cover”. Next we want to make sure you have the Insert Tool¹. Press on the + in the lower left, then Applications then Go to App Store.



Now in the app store, type in Insert Tool and press Enter.

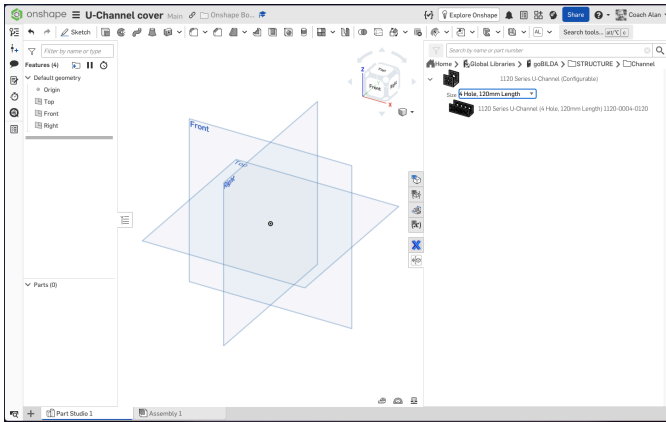


If yours doesn’t say “Subscribed” then click on it and subscribe.

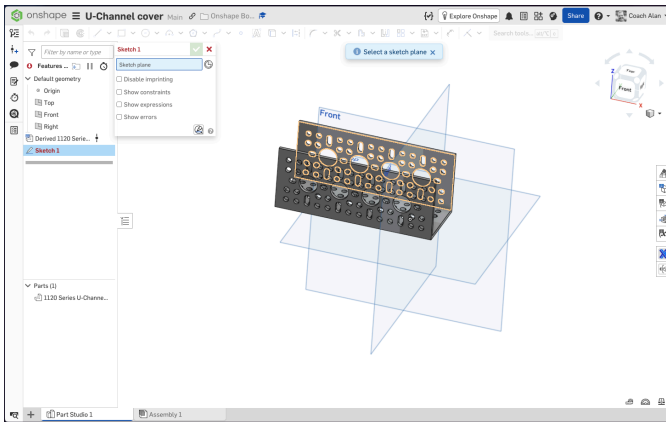
Click on the Insert tool, Then Home -> Global Libraries -> goBILDA -> STRUC-TURE -> Channel. Select the 1120 Series U-Channel (Configurable) and change its size to 4 Hole. It should now look like this:

¹ Provided by Cardinal Gibbons Robotics FTC Team 2901

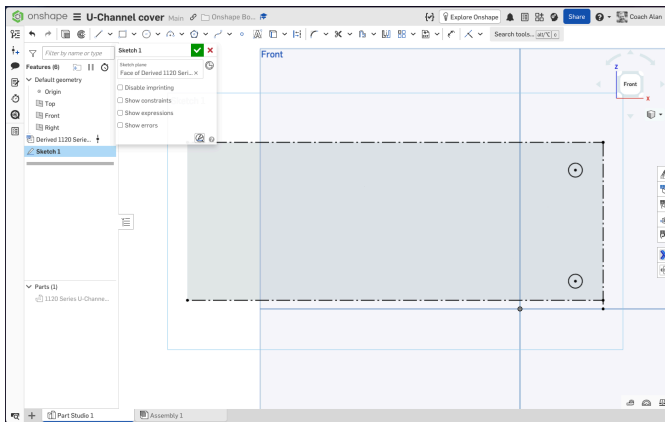
7. Cover end of U-Channel



When we click on the channel, it will “derive” into our part studio. We can then use it in our sketches. (If you are in an assembly, then you can insert it just like your own custom parts. We’ll do that starting in [Part III.](#)) . Click on the insert tool again to hide it and now lets create a sketch on the inner wall of the U-Channel. (**Shift+S**).

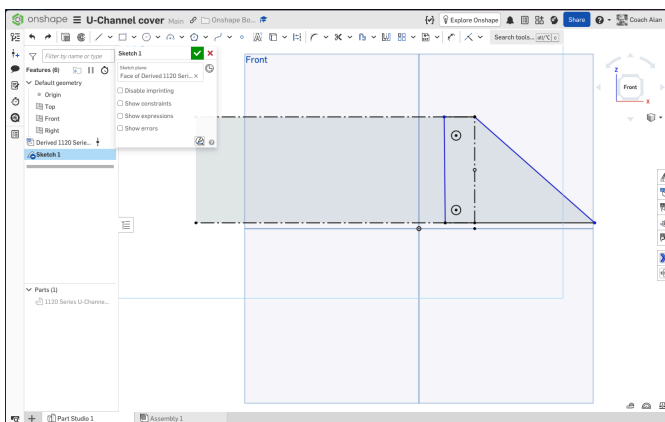


Much like the last chapter, we are going to “use” the holes we want it to go through. So select “use” (**U**) and click on the two holes. It will help us to also have construction lines showing the top bottom and edge of the U-channel so select “construction line” (**Q**) and click on the top, bottom, and right edge of the u-channel. Now go hide the derived part (select under parts and click on the eye). Your sketch should now look like:



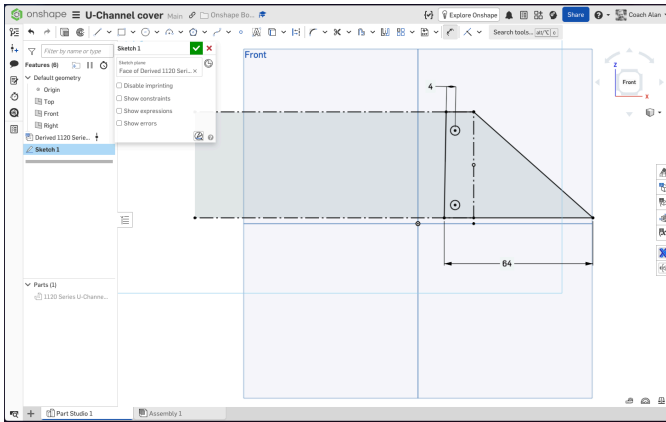
If you have a gap between the gray and your bottom line, it probably means that you selected the outside of the material inside of the edge on the inside.

Now use the Line tool () to draw four lines.

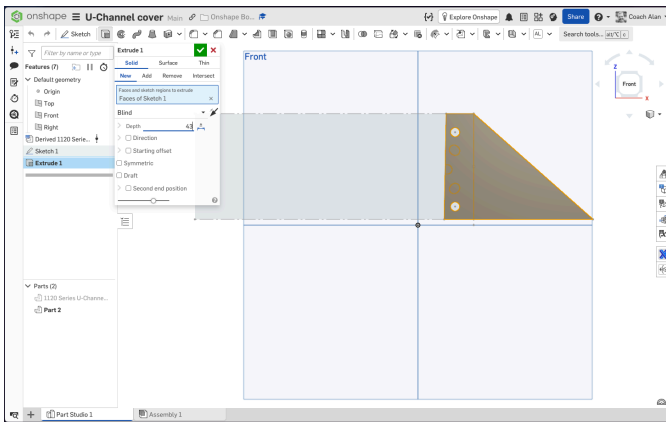


Now select Dimension () and make the vertical line 4mm away from the center of the hole. (I chose 4mm because there are 8mm between holes on the Gobilda pattern so it is half. But you could choose a bigger number if you felt like there wasn't enough material here.) . Then make the length of the long line 64mm. Your sketch should now look like this (I moved the 4mm outside of the arrows so I could read it.)

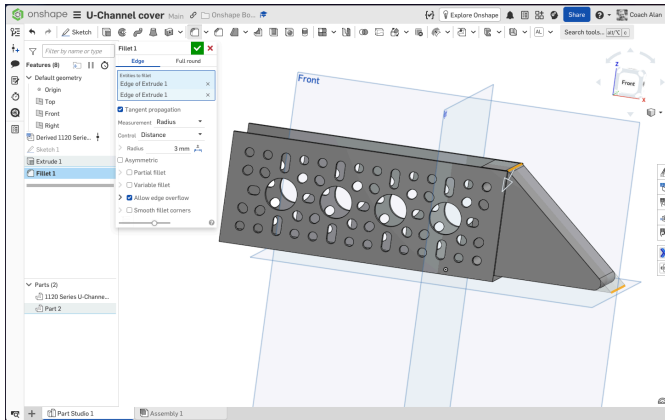
7. Cover end of U-Channel



Now let's Extrude (**Shift+E**). We are going to make the depth 43mm. (You can look that up on GoBilda's website to see that is the empty amount of space between walls in U-Channel.) and accept it.



You can show the U-Channel again and it is pretty obvious we have some sharp edges on our part that are exposed. This time instead of using a sketch fillet we'll just use a normal fillet. Choose the fillet item (press **Shift+F**) and then click on the two sharp edges. (You can do all edges, but I like to not do them on the surface I am 3d printing on because then you have some unsupported area on your print.)



To 3d print this part, just export Part 2. (Yes, you probably want to rename it.)

This is one of three ways to connect a part. In it we make a hole all the way through the part and we need a long screw to go through it. This works and is often a pretty easy way to do it. However there are two other ways.

1. Make the part slightly smaller than a screw and have the screw “bite” into the plastic. This can be a win for easy, but be aware that it limits how many times you can remove and reinsert the part before the screw doesn’t hold anymore. This would involve changing the size of the hole from 4.2 to perhaps 3.8mm for the diameter.
2. Use brass inserts. If you aren’t familiar with brass inserts, they are very cool. They allow you to put a set of screw threads into your 3d printed part. This allows you to screw into it just as if it was tapped. The ones we typically use (M4x4), we set a diameter hole of 5.6mm and if it isn’t a through hole we make it 5mm deep. Then you put the brass insert in the hole with a soldering iron (heated) on the inside and press on the brass insert until it sinks into the part.

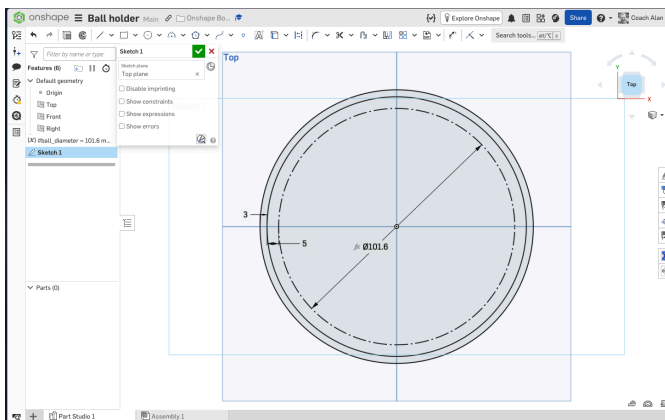
8. Ball holder

Several years FIRST has had game pieces that are spherical. This is actually great because they are relatively easier to design mechanisms for as they are the same no matter how they are rotated.

So let's imagine that we are trying to make a basket to carry a 4" diameter ball in. While we could just make it a square basket close to the size or a cylinder, we will use this opportunity to talk about revolving sketches.

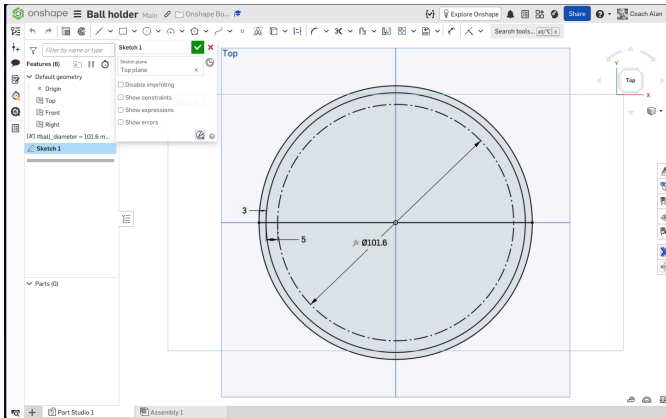
Create a new document called "Ball holder" and make a sketch on the Top plane (**Shift+S**) and press **N** so we are looking at it top down.

Start by making a circle the diameter of the ball as a construction line. (**C** for center point circle, **Q** for construction line). Put in #ball_diameter as the diameter and define it as a new variable that is 4in. We then want a circle (not construction line, so press **Q** to turn off construction line) that is offset from it (**O** for offset) by 5mm (to give some tolerance) and then another circle with an offset of 3mm (thickness of shell we want). It should look like this (I moved the offset amounts to make them easier to read)

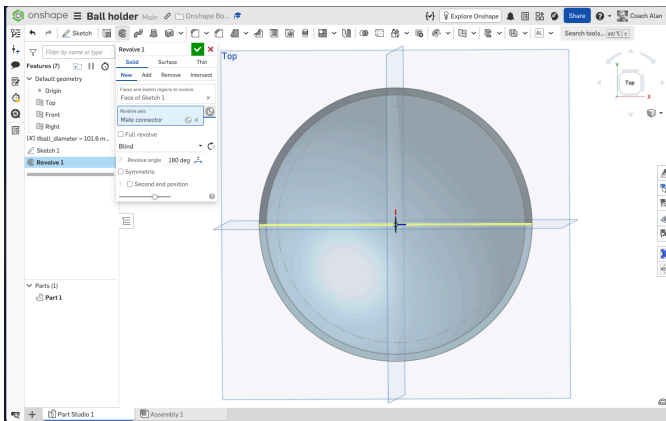


Because a revolve will take a part of our sketch and revolve it around an axis, we need a line dividing the circle in half. So let's make a horizontal line from one edge of the circles to the other. (Select Line or **L**) and you'll end up with a sketch like this:

8. Ball holder

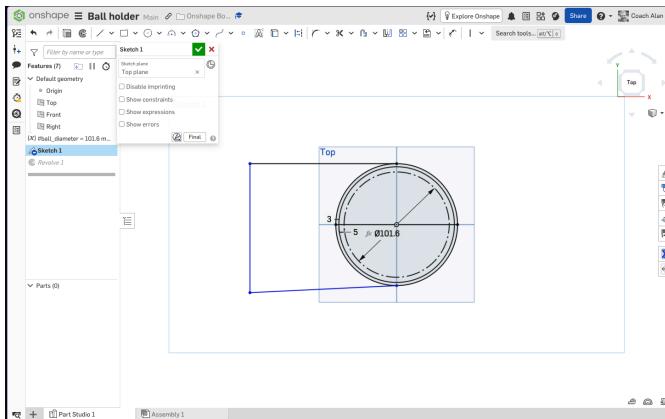


We select Revolve (**Shift+W**¹), select the bottom half of our circle and select the axis as our mate connector. Note that we change from a full revolve to having a revolve angle of 180 (because we don't want a sphere, but only half of one.) . It should look like this:

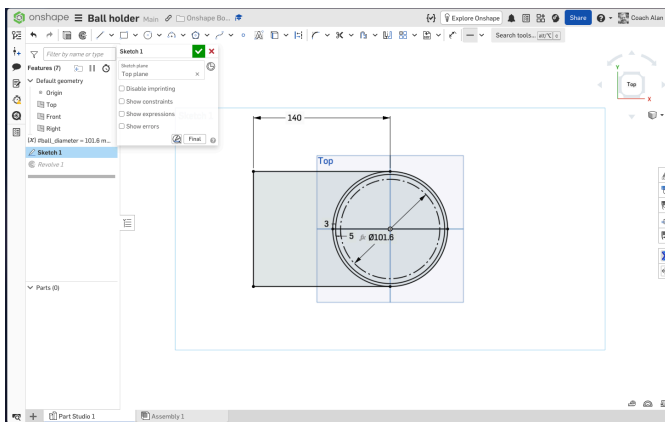


Select the green check to accept. This is pretty cool, but we still need a way to attach it to our robot. So right click on Sketch 1 and select “Edit...” (if you played with rotating it around to see it, you'll want to press **N** to get it back to being top down. Let's draw 3 lines.

¹I have no idea why the default keyboard shortcut for revolve is **Shift+W**

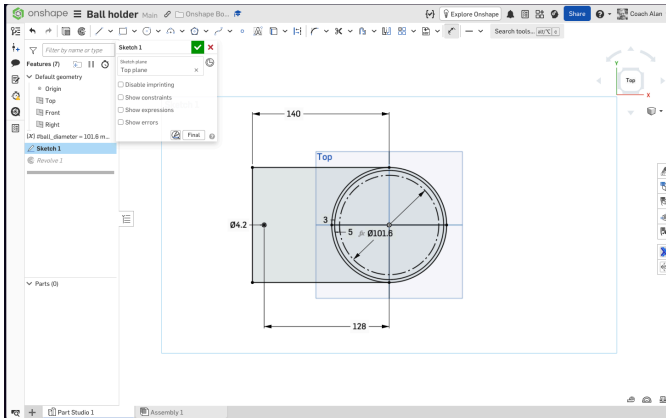


Let's apply the horizontal constraint (**H**) to the bottom line and a dimension of 140mm to the top line. For mine this made everything black (constrained.) . You may have to work with the various constraints to get yours all black. This is what it should look like:

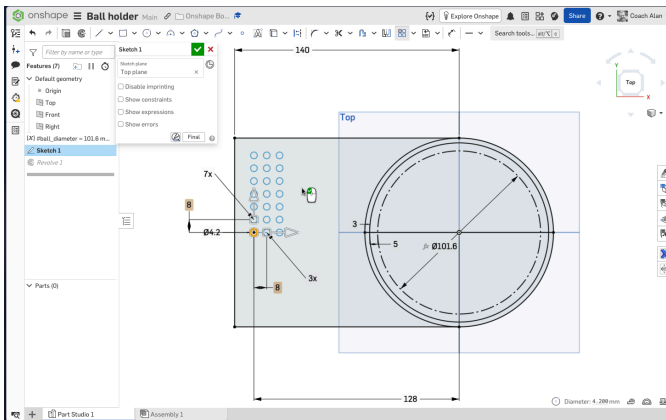


We need some holes on here to make it easy to attach. So select center point circle (**C**) and give it a size of 4.2mm and put it on the axis. Then give a dimension of 128 from the center of our new circle to the origin. It should look like:

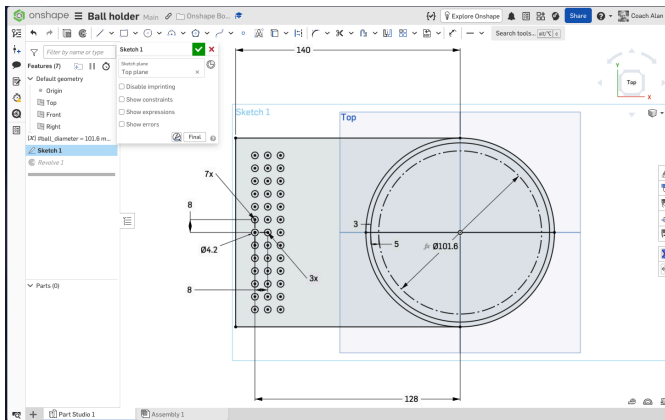
8. Ball holder



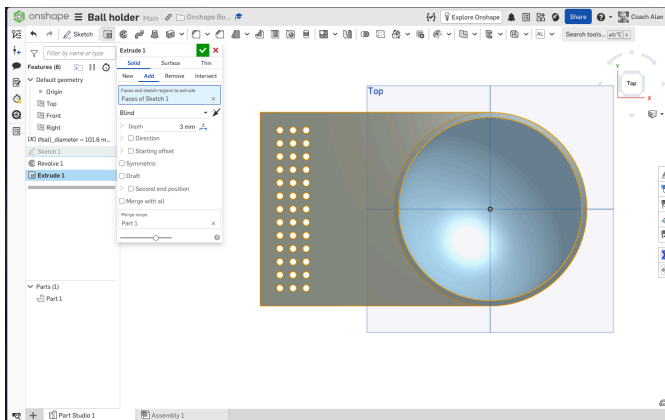
Now we are going to make a field of holes so we'll have lots of choices. Remember that GoBilda is on an 8mm pattern. We are going to use one of my favorite tools which is sketch pattern to show how to make a field of holes. Select Linear Pattern and then click on our circle. Change the vertical from 1x to 7x and the horizontal from 1x to 3x and change both of the amounts from 25 to 8mm. It should look like this (I moved the 8mm so you could read the number)



Press outside of the drawing area with the left mouse button to accept. Now select mirror and select the horizontal axis as your mirror line and use the select rectangle to select all of the holes (except the row on the axis). You'll get something like this: (I moved the 8mm and the diameter of the hole measurement so we could see them)



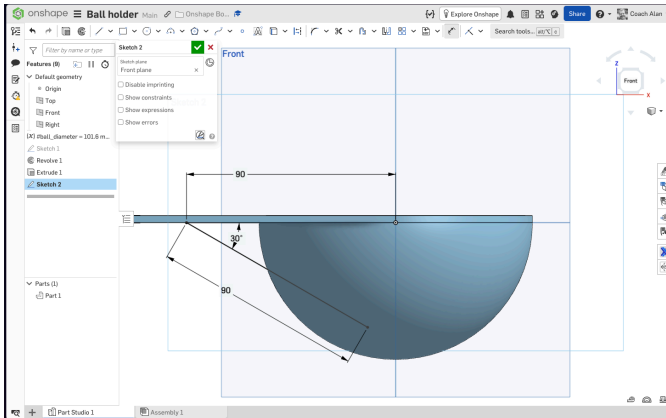
Now let's extrude that by 3mm (**Shift+E**) and add it to our part.



Now you might look at this and say that you are afraid it will break where the ball catcher piece meets the flat piece. It would be good if we could add a rib to connect those two.

To do this. Let's create a new sketch (**Shift+S**) on the Front plane and hit **N** to make it so we are looking top down. Let's draw two lines on this sketch even though we only need one because it makes it easier to constrain. You'll notice that I made a line 90mm long coming out from the origin on the axis, and then I made a line that was also 90mm long (It doesn't really matter, it just needs to go significantly into the sphere.) . I made the second line come out at a 30 degree angle from the first. To do this, select Dimension (**D**) and click on both lines and enter 30. It should look like this:

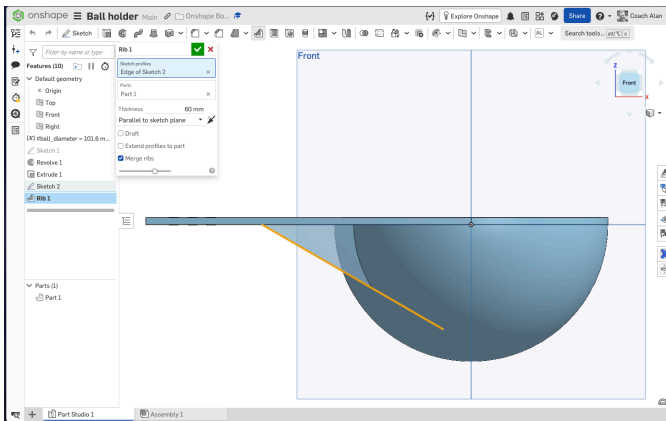
8. Ball holder



Now accept the sketch (clicking on the green check mark)

Select Rib and click on your line for the sketch profile and Part 1 for the parts. Change the Thickness to 60 (I am completely making that number up.) . You'll have to change the arrow next to Parallel to sketch plane if your rib ends up inside your sphere.

You should now have a part you can be proud of that looks like this:

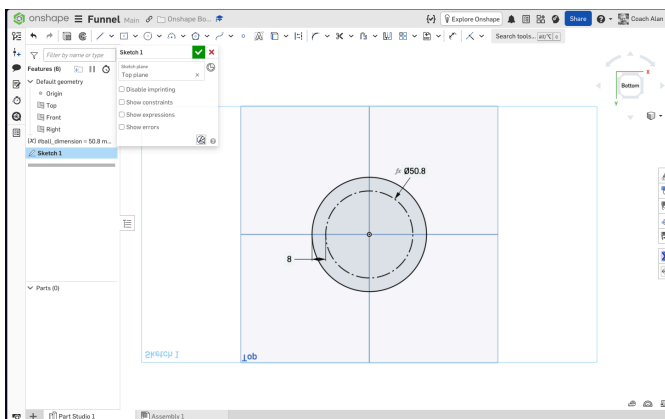


This is a great example where you probably want to apply fillets to make the part stronger but you can't in a sketch because the edges weren't created there. Select Fillet (**Shift+F**) and then select the three faces of the rib. (You could also select all of the edges individually but I think this is easier) . It will look like:

9. Funnel

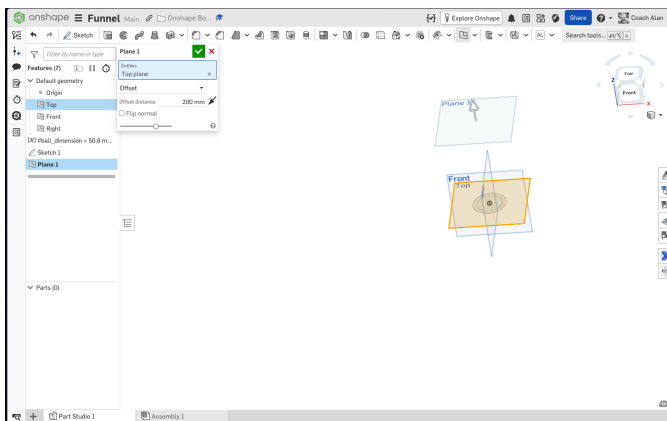
Let's imagine that we want a way to create a funnel. Perhaps a game is all about catching balls but we want them to end up in a certain place. For this particular game the balls are 2 inches in diameter. We will do just like we did in [chapter 8](#) where we make a circle with a construction line the size of the ball and then we'll use offset to make a larger circle. (In this case we do 8mm because we just need the outside of the material. This one is 5mm for tolerance + 3mm for wall thickness.)

It should look like this:

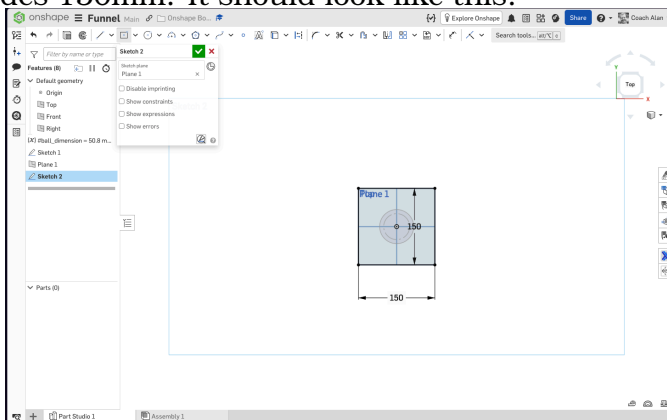


Accept it. Now we are going to create a new plane that is offset 200mm from the top plane. If you zoom it out, you can see it as we create it.

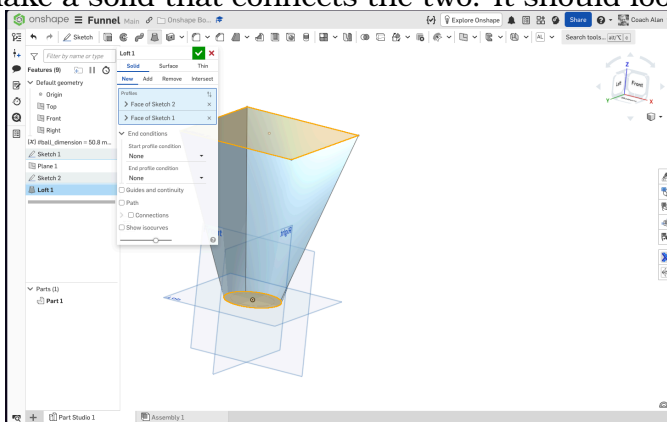
9. Funnel



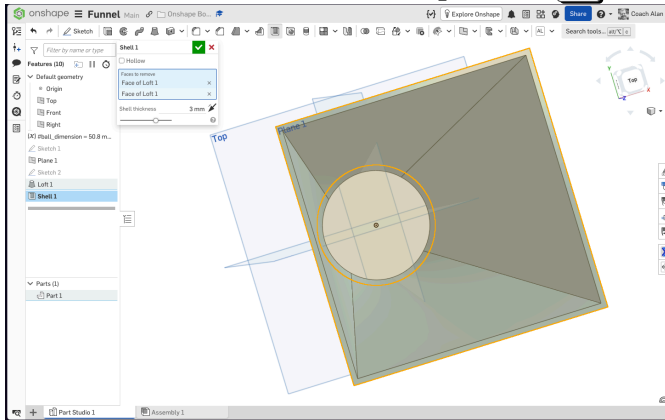
Accept it (green checkmark) then Create a sketch (**Shift+S**) selecting our new Plane 1 as the Sketch Plane. Create a Center Point Rectangle (**R**) with both sides 150mm. It should look like this:



Accept it and then select Loft selecting the face of the two sketches. This will make a solid that connects the two. It should look like this:



Accept it (green checkmark). Well, we need it to be hollow to be of any use. So select Shell and select both the top and bottom face of the loft. It should look like this (I turned it around and pressed **F** to zoom to fit.):



Once again, we need some way to attach this to our robot. But in this case, we'll get you to try it yourself using what we did at the end of [chapter 8](#). You could make it at the top or bottom of the funnel (or both)

Part III.

Modeling your robot

This section is not ready for review yet

10. Design your robot - Space Planning

For this book, we'll design a relatively simple robot. While we often will iterate on design as we learn more, I recommend that we start with an idea and do rough space planning. This allows multiple people on the team to work on CAD at the same time.

Hint: - make the origin the center (x, y) of all available robot space at starting. So for FTC, you make it the center of the 18 inch cube allowed. The goal here is to do something rough quickly.

10.1. Space planning for Drive train

1. Make a sketch in a part studio
2. Make a center point rectangle that is "18in x 18in" with two midpoint lines. Make them all construction lines
3. Make our left channel + wheel space rectangle (440x94) that is 132 from the center. Make a point, constrained midpoint. Make the point coincident with construction line.
4. Use Mirror to make the right one.
5. Make the back piece (48mm, entire width). 32 mm from the back.
6. Make a center connecting piece (48mm, entire width), centered on center of robot

A piece of advice we got from a mechanical engineering mentor was that it should be clear which mechanism gets to dictate for other mechanisms. For example: the shooter might state where the ball has to be on the robot to be shot and then the intake is responsible for designing it to be delivered there.

10.2. Space planning for intake

(Go through this and realize that the drive train can't be as big as we made it so reduce it)

10.3. Space planning for electronics and battery

11. Mecanum Drivetrain

Start with design. While the standard GoBilda chassis is pretty good, it can make it difficult to assemble your own mechanisms because of all of the space motors take up inside the chassis.

So we'll start with a mecanum chassis that has the motors sticking up so we have the maximum amount of space for adding things.

Why would we use CAD to put together standard parts instead of just building it? A few reasons:

1. It turns out that it is much faster to use CAD to put the parts together
2. It allows you to figure out what parts you will need when you might not have all of them.
3. It allows you to find problems earlier

12. Intake

Just like we did earlier, we need to make sure we know what we are designing before we start.

Put together an intake that is a motor with a shaft with gecko wheels on it.

13. Bringing it all together

Appendices

A. List of Keyboard Shortcuts mentioned

You can see a full list by clicking on the ? icon in the top right and selecting Keyboard Shortcuts. But here I have listed the ones I mention in the book. (The most common ones I use.)

A.1. In 3D viewing

F	Zoom to Fit
N	Set to “Normal” - perpendicular
P	Show/Hide Planes

A.2. In Part Studio

Shift+F	Fillet
Shift+S	Create Sketch

A.3. In a Sketch

C	Center point Circle
D	Dimension
Shift+E	Extrude
Shift+F	Fillet
H	Horizontal Constraint
Shift+M	Midpoint Constraint
O	Offset

A. List of Keyboard Shortcuts mentioned

Q	Construction Line
R	Center point Rectangle
U	Use (convert/project)
Shift+W	Revolve

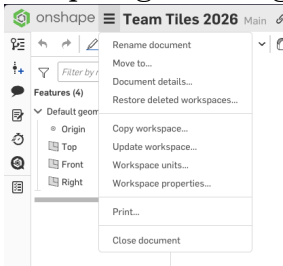
A.4. In an Assembly

m	Add a Fastened mate
---	---------------------

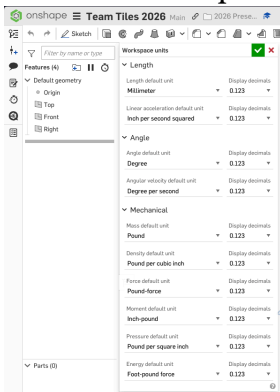
B. Setting Workspace Units

If someone else created the documents and it is set to inches, you can change the document to mm (millimeters).

To do this, click on the “hamburger” menu ($\equiv$) in the upper left after the Onshape logo. You’ll get something like this:



Click on Workspace units and you’ll get something like this:



You could argue (and I would agree with you) that you should change all of the units to be metric. But I typically only change the other defaults when I need to use them since I don’t use them in many documents.

C. Configurations

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